TS5000 Robot Controller

Fieldbus Slave Manual

Rev. 12

December 6, 2023

Notice

- Make sure that this instruction manual is delivered to the final user of Shibaura Machine's industrial robot.
- Be sure to read this manual and that you fully understand its contents before operating the industrial robot.
- After reading through this manual, keep it in an easily accessible place for future reference.

Shibaura Machine

This instruction manual is intended for the TS5000 controller.

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Introduction

Thank you for purchasing this Shibaura Machine robot system.
This manual describes the initial setting procedures and operating procedures for the Fieldbus slave function of the TS5000 controller.
This manual is intended for users who have a basic knowledge of Fieldbus.
The Fieldbus slave module can use CC-Link, EtherNet/IP, PROFIBUS, DeviceNet, PROFINET, or EtherCAT.
The Modbus TCP Server function can be used as a Fieldbus only when the slave module for the Fieldbus is not used.

The Fieldbus input signals are assigned to DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, and DIN1001 to DIN1064, and the output signals are assigned to DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, and DOUT1001 to DOUT1064. (The available bits and the data range vary depending on the Fieldbus type and settings.)
Also, TCmini (internal PLC) can be used to change the I/O signals and to input and output system signals over Fieldbus.

Manufacturer

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If you have purchased this controller outside of Japan, please contact the company that you purchased it from.

Table of Contents

Introduction

Table of Contents

Scope of Use

Usage Procedures

Warranty

Standards, laws, and regulations pertaining to the safety of industrial robots

Safety Information

Service Network

Specifications

1. Fieldbus Common Settings 22
   - 1.1 Parameters 22
     - [P009] FIELDBUS COMMON SETTING 22
2. Ethernet/IP Slave Module 24
   - 2.1 Specifications 24
   - 2.2 Hardware Configuration 25
     - 2.2.1 LED Screen 25
     - 2.2.2 Connector Pin Arrangement 26
     - 2.2.3 Cables 26
   - 2.3 Parameters 27
     - [P010] ETHERNET/IP SETTING 27
3. CC-Link Slave Module 29
   - 3.1 Specifications 29
   - 3.2 Hardware Configuration 30
     - 3.2.1 LED Screen 30
     - 3.2.2 Connector Pin Arrangement 30
     - 3.2.3 Cables 30
   - 3.3 Parameters 31

[P011] CC-Link SETTING 31

4. PROFIBUS Slave Module 33

4.1 Specifications 33
4.2 Hardware Configuration 34
4.2.1 LED Screen 34
4.2.2 Connector Pin Arrangement 35
4.2.3 Cables 35
4.3 Parameters 36
[P012] PROFIBUS SETTING 36

5. DeviceNet Slave Module 37

5.1 Specifications 37
5.2 Hardware Configuration 38
5.2.1 LED Screen 38
5.2.2 Connector Pin Arrangement 38
5.2.3 Cables 39
5.3 Parameters 40
[P013] DEVICENET SETTING 40

6. PROFINET Slave Module 41

6.1 Specifications 41
6.2 Hardware Configuration 42
6.2.1 LED Screen 42
6.2.2 Connector Pin Arrangement 43
6.2.3 Cables 43
6.3 Parameters 44
[P014] PROFINET SETTING 44

7. EtherCAT Slave Module 46

7.1 Specifications 46
7.2 Hardware Configuration 48
7.2.1 LED Screen 48
7.2.2 Connector Pin Arrangement 49
7.2.3 Cables 49

8. Modbus TCP Server 50

8.1 Specifications 50
8.2 Hardware Configuration 51
8.2.1 Cable 51
8.3 Parameters 51
[P016] MODBUS PROTCOL SELECT 51
8.4 Communication Specification 52

8.4.1 Communication Protocol 52
8.4.2 Function codes 52
8.4.3 Addresses 53
8.4.4 Communication format of Each Function Code 54
0x01, 0x02 55
0x03, 0x04 56
0x05 57
0x06 59
0x0F 60
0x10 61
9. Alarm Detection 63
10. Input/Output Addresses 64
11. TCmini (Internal PLC) 65
11.1 Bit Input (FieldBUS→TCmini) 65
11.2 Bit Output (TCmini→FieldBUS) 66
11.3 Data Input (FieldBUS→TCmini) 67
11.4 Data Output (TCmini→FieldBUS) 67

Operation

12. Operation Verification 69
13. Fieldbus System Signal Input/Output Function 73
13.1 System Input Signal Assignment Configuration 74
13.1.1 System Configuration at Function Disabled (SEQ-FUNCTION-SW1= 0) 74
13.1.2 System Configuration at Function Enabled (SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3 = 0) 75
13.1.3 System Configuration at Function Enabled (SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3= 1) 76
13.2 System Output Signal Assignment Configuration 77
13.2.1 System Configuration at Function Disabled (SEQ-FUNCTION-SW1= 0) 77
13.2.2 System Configuration at Function Enabled (SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3 = 0) 78
13.2.3 System Configuration at Function Enabled (SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3= 1) 79
14. TCmini (Internal PLC) Function (Sample Connection Example) 80
14.1 Operating the Robot from an Upper-level PLC 81
14.2 Sending Robot Position Information to Master Device 84
14.3 Sending Teaching Points from an Upper-level PLC to Robot 86

15. Operations Using SCOL2 Language 88
15.1 Sample Program 89
16. Verifying the Module Version 90
16.1 Verifying the Version of the Fieldbus by Using TSAssist 90
16.2 Verifying the Version of the Fieldbus by using TP5000 92

Scope of Use

The product has been developed for use in a wide range of automation tasks including transport work such as the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This product is not intended for use in tasks involving human contact (such as tasks for medical care and nursing care) or to be incorporated in equipment that sustains human life (such as life support devices and incubators), equipment on which people ride (including transportation equipment such as cars and aerospace equipment such as airplanes, and recreational equipment such as roller coasters), and equipment that protects human life or the human body (such as safety devices).
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used in this way.

Usage Procedures

This section describes the usage requirements for the TS5000. Usage of the product in a state outside the required conditions could result in unexpected risks. Be sure to always use the controller in an environment that meets the required conditions.

1. Main Specifications of Product: Refer to the robot manual for your robot.
2. Usage Restrictions: Tables 1 to 6 provide the "Restricted Specifications Sheet" indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Clear a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Clear the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the fixing bolts of the robot fixing jig. [5] Use a crane or cart to transfer the robot to the installation site. - Controller [1] Use a cart to transfer the controller to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Press the robot by hand to move to a location where it is easy to perform work. [6] Mount the end effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [6] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jigs can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jigs. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] If using a crane to transfer the robot, mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Clear a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the greasing state and apply grease if needed (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [6] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jigs can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jigs. [4] Turn off the power, and remove the wiring. [5] If using a crane to transfer the robot, mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot and controller using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or where excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power while these covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that a stop to wait for an operation signal during operation was an operation stop - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be provided. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and movable range of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cables shall be installed at a location where they will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided so that, if someone enters the hazard zone, they will not get entrapped by the robot or peripheral equipment. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the robot hand, mount the fixing jigs, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box.

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restrictions on space)		
2) Interface (continued)	Maintenance: Disposal:	Space required for maintenance shall be provided. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided so that, in the hazard zone, a person will not get entrapped by the robot or peripheral equipment. Work shall be able to be performed to remove the robot hand, mount the fixing jigs, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the area during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be generated.	

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, the risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal”.

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	-
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot inspection work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period

The warranty period of the product is one of the following, whichever is shorter.

(1) 18 months after shipping from our factory
(2) 12 months after installation at customer site

2. Warranty scope

(1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
(2) We will repair a faulty product free of charge within the warranty period only if this product has been operated and used in accordance with the instruction manual supplied with the product. A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimers

The following cases are outside the scope of the warranty.

(1) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
(2) Any failure caused by aging or normal wear and tear (natural fading of paints, deterioration of consumable parts[1)].
(3) Irregularities confined to sensory quality such as noise not recognized as affecting functions
(4) When any modification or disassembly without our consent was performed
(5) Any failure or damage arising from inadequate maintenance and inspection or improper repair
(6) Any failure or damage due to an act of God, fire, and other force majeure
(7) Internal data such as programs and points created or modified by customers
(8) When this product was purchased in Japan and taken outside the country

*1) Consumable parts refer to the replacement parts for maintenance as listed in the manual of the controller.

4. Safety information

(1) Shibaura Machine shall provide no guarantees for the standard performance of this product when used outside the specifications.
(2) If the customer does not observe the safety information appearing for DANGER, WARNING, CAUTION, and NOTES contained in this instruction manual, Shibaura Machine will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.
(3) Note that the safety information appearing for DANGER, WARNING, CAUTION, NOTES, and other explanations in this instruction manual are within the scope of usage assumed by Shibaura Machine, and they do not foresee all possible incidents for all situations that could occur.

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.

Based on the above, the ISO/IEC has defined "Guide 51", which classifies the various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.

- ISO 10218, ISO 10218-1

Be sure to always follow the standards, laws, and regulations set forth in the country where the industrial robot is used.

Safety Information

This instruction manual contains important information to ensure the safe and correct use of the controller and to prevent injury to the operator and others and damage to property.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

Display of Signal Words and Their Meanings

Signal Word and Symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to the operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2)] only and provides helpful advice about performing work.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

⚠ DANGER
• In order to ensure controller safety from transportation to disposal, be sure to carefully read the safety manual shown in the target controller manual. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts

Specifications

This section describes the Fieldbus slave specifications of the controller.

1. Fieldbus Common Settings

1.1 Parameters

The Fieldbus common settings are configured using P009 in the PLC.PAR file.

For details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

[P009] FIELDBUS COMMON SETTING

Parameter key	FBUS-ENABLE
Parameter data	= (Fieldbus communication)
Data type	Integer type
Data count	1
Data range	-
Description	This parameter is used to enable/disable Fieldbus communication. When Disable is selected, Fieldbus communication is not performed.
	0: Disable Fieldbus communication 1: Enable Fieldbus communication
Default	1

Setting example:

When disabling Fieldbus communication

```
‘[P009] FIELDBUS COMMON SETTING
FBUS-ENABLE = 0
```

Parameter key	FBUS-ERROR-JOG-ENABLE
Parameter data	= (Enable/Disable Fieldbus alarm)
Data type	Integer type
Data count	1
Data range	-
Description	In the Fieldbus settings, servo ON/OFF control is disabled when an alarm is detected, and so this parameter temporarily disables the Fieldbus alarm. *The alarm can be disabled by this parameter in MANUAL mode only. When in AUTO mode, the alarm will not be disabled even if this parameter is set to 1.
	0: Show Fieldbus alarm 1: Disable alarm
Default	0

Setting example:

When disabling the Fieldbus alarm in MANUAL mode

```
‘[P009] FIELDBUS COMMON SETTING
FBUS-ERROR-JOG-ENABLE = 1
```

Parameter key	FIELDBUS-ENDIAN
Parameter data	= (Fieldbus endian)
Data type	Integer type
Data count	1
Data range	-
Description	Set the endian of the Fieldbus master device. The controller communicates with the Fieldbus master by converting the bit fields to those corresponding to the endian of the Fieldbus master.
	0 ：16bit big endian (0x1234 5678→0x3412 7856) 1 ：16bit little endian (0x1234 5678→0x1234 5678) 2 ：32bit big endian (0x1234 5678→0x7856 3412) 3 ：32bit little endian (0x1234 5678→0x5678 1234)
Default	1

Setting example:

The Fieldbus master device communicates in 16-bit little endian mode.

```
‘[P009] FIELDBUS COMMON SETTING
FIELDBUS-ENDIAN = 1
```

2. Ethernet/IP Slave Module

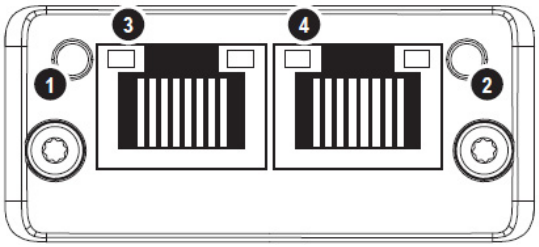
2.1 Specifications

Supported Fieldbus specifications	EtherNet/IP
Module model	TS5K-IP
Module type	Scanner
Supported master device	KV-7500 by Keyence
EDS file	Each device over an Ethernet/IP network is associated with an EDS file that contains all required information for the device. The EDS file can be used under the network configuration. This file can be loaded to make the basic settings. **The EDS file is included in the Instruction Manual CD.**
Number of inputs	512 inputs DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, DIN1001 to DIN1064
Number of outputs	512 outputs DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, DOUT1001 to DOUT1064
Input data count	16 words D700 to D70F
Output data count	16 words D740 to D74F
Supported software version	Main unit: X8KCS-02A or later

2.2 Hardware Configuration

2.2.1 LED Screen

Front View

#	Item	Connector	
1	Network status LED	Ethernet, RJ45	
2	Module status LED		
3	Link/Activity LED (Port 1)		
4	Link/Activity LED (Port 2)		

Network status LED

LED status	Description
Off	Power off or no IP address
Solid green	Online or one or more connections are established (CIP class 1 or 3)
Blinking green	Online or connection not established
Solid red	Duplicate IP address or critical error
Blinking red	One or more connections have timed out (CIP class 1 or 3)

Module status LED

LED description	Description
Off	Power off
Solid green	Controlled by scanner in Run status
Blinking green	Not set or scanner is idle
Solid red	Major fault (exception state, critical error, etc.)
Blinking red	Recoverable fault

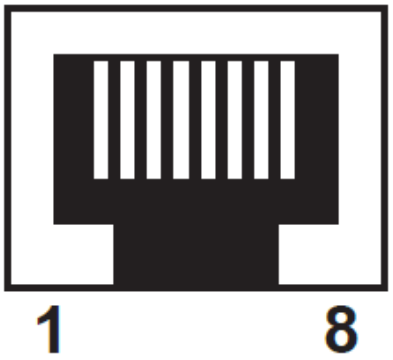
Link/Activity LED 3/4

LED status	Description
Off	No link, no activity
Solid green	Link (100 Mbps) established
Blinking green	Activity (100 Mbps)
Solid yellow	Link (10 Mbps) established
Blinking yellow	Activity (10 Mbps)

2.2.2 Connector Pin Arrangement

Same for Port 1 and Port 2

Pin	Signal name
1	TXD+
2	TXD-
3	RXD+
4	-
5	-
6	RXD-
7	-
8	-



2.2.3 Cables

For the cables used for the external connections, use the cables recommended for the master device that you use.

RJ45 connector: Cat 5 shielded cable (straight cable)

2.3 Parameters

The Ethernet/IP settings are configured using P010 in the PLC.PAR file.

For details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

[P010] ETHERNET/IP SETTING

Parameter key	ETHERNETIP-ADD
Parameter data	= (IP address)
Data type	Integer type
Data count	4
Data range	-
Description	- The IP address is an address for identifying the Ethernet connection device. - The address is expressed using four numbers from 0 to 255 separated by a comma (,). - If a PC connected to the Ethernet/IP and network is connected directly using a 1-to-1 connection, the initial value (any selected) can be used for the IP address. However, if the PC is connected to an existing LAN, set the IP address based on instructions from your LAN system administrator. If IP addresses are duplicated, the functions may not work correctly. - The PC or PLC that communicates with the Ethernet/IP must be connected to the same network.
Default	= 192,168,0,1

Parameter key	ETHERNETIP-MASK
Parameter data	= (Subnet mask)
Data type	Integer type
Data count	4
Data range	-
Description	This specifies the subnet mask of the connected network. This must be set when connecting to an existing LAN. Set the subnet mask based on instructions from your LAN system administrator.
Default	= 255,255,255,0

Parameter key	ETHERNETIP-GATEWAY
Parameter data	= (Gateway)
Data type	Integer type
Data count	4
Data range	-
Description	This specifies the default gateway address. This must be set when connecting to an existing LAN. Set the default gateway address based on instructions from your LAN system administrator.
Default	= 0.0,0.0

Setting example:

When EtherNet/IP is selected and the following is set in the KV-7500 made by Keyence.

IP address: 192.168.10.1

Subnet mask: 255.255.255.0

Gateway: 0.0.0.0

```
'[P010] ETHERNET/IP SETTING
'{IPAdress}
ETHERNETIP-ADD = 192, 168, 10, 10
'{Net Mask}
ETHERNETIP-MASK = 255, 255, 255, 0
'{GateWay}
ETHERNETIP-GATEWAY = 0, 0, 0, 0
```

3. CC-Link Slave Module

3.1 Specifications

Supported Fieldbus specifications	CC-Link Ver. 1.10/Ver. 2		
Module model	TS5K-CC		
Module type	Remote device station		
Supported master device	KV-7500 by Keyence		
Node address	The node address is specified by the PLC parameters. (1 to 64)		
Communication speed	The communication speed is specified by the user parameters. 156 kbps / 625 bps / 2.5 Mbps / 5 Mbps / 10 Mbps		
CC-Link version	Ver. 1	Ver. 2	
Number of stations used	4		
Expansion cyclic settings	1x	1x	2x
Number of inputs	128[1)] DIN301 to DIN364, DIN401 to DIN464	128[1)] DIN301 to DIN364, DIN401 to DIN464	224[1)] DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN632
Number of outputs	128[1)] DOUT301 to DOUT364, DOUT401 to DOUT464	128[1)] DOUT301 to DOUT364, DOUT401 to DOUT464	224[1)] DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT632
Input data count	16 words D700 to D70F	16 words D700 to D70F	32 words D700 to D71F
Output data count	16 words D740 to D74F	16 words D740 to D74F	32 words D740 to D75F
Supported software version	Main unit: X8KCS-02A or later		

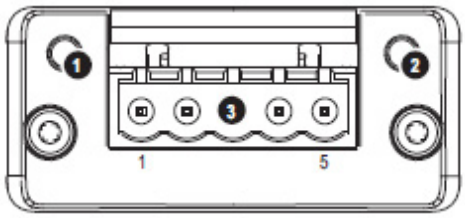
1) The last 16 inputs/outputs are used as a system area, and so they cannot be used by the customer.
In the CC-Link settings, select the options for four stations used and remote device station.

3.2 Hardware Configuration

3.2.1 LED Screen

Front View

#	Item
1	Run LED
2	Error LED
3	CC-Link connector



Run LED

LED status	Description
Off	Not joined the network, timeout status (power is turned off)
Solid green	Joined the network, normal operation
Solid red	Severe fault (Fatal error)

Error LED

LED status	Description
Off	No error was detected (power is turned off)
Solid red	Severe fault (exception or fatal event)
Flickering red	CRC error (temporary flickering)
Blinking red	Station number or baud rate was changed after startup (blinking)

3.2.2 Connector Pin Arrangement

Pin	Signal name	Application
1	DA	RS485 RxD/TxD (DA)
2	DB	RS485 RxD/TxD (DB)
3	DG	Signal ground (DG)
4	SLD	Cable shielding
5	FG	Protective ground

Fieldbus connector: Terminal block is used

3.2.3 Cables

For the cables used for the external connections, use the cables recommended for the master device that you use.

Model: V1.25-M3 or equivalent

3.3 Parameters

The CC-Link settings are configured using P011 in the PLC.PAR file.

For details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

[P011] CC-Link SETTING

Parameter key	CCLINK-NODE
Parameter data	= (Node)
Data type	Integer type
Data count	1
Data range	0 to 64
Description	This sets the Fieldbus node address. Set the node address so that it does not duplicate the address of any other devices on the same network. The range of setting values varies depending on the Fieldbus type.
	0: No Fieldbus options 1 to 64: Node address valid range in CC-Link
Default	1

Parameter key	CCLINK-SPEED
Parameter data	= (Communication speed)
Data type	Integer type
Data count	1
Data range	0 to 4
Description	This sets the communication speed. Set it to match the communication speed of the master device.
	0: 156 kbps 1: 625 kbps 2: 2.5 Mbps 3: 5 Mbps 4: 10 Mbps
Default	0

Parameter key	CCLINK-MODE
Parameter data	= (setting for number of inputs/outputs)
Data type	Integer type
Data count	1
Data range	0 to 2
Description	This sets the number of inputs and outputs used by CC-Link. 0: Inputs and outputs: 128 each (16 bytes), 16 words (initial value) = 0 Inputs and outputs: 128 each (16 bytes), 16 words (TS3000 compatible, CC-Link Ver. 1) = 1 Inputs and outputs: 128 each (16 bytes), 16 words (4 stations, 1x: CC-Link Ver. 2) = 2 Inputs and outputs: 224 each (28 bytes), 32 words (4 stations, 2x: CC-Link Ver. 2)
Default	0

Setting example:

Settings for node address: 1, communication speed: 156 kbps, inputs and outputs: 224 each (28 bytes), 32 words

```
‘[P011]  CC-Link SETTING
‘{Node Addr}
‘{0:DISABLE, 1-64: ENABLE}
CCLINK-NODE = 1
‘{Speed}
CCLINK-SPEED = 0
‘{Mode}
CCLINK-MODE = 2
```

4. PROFIBUS Slave Module

4.1 Specifications

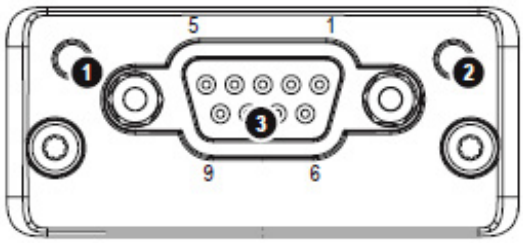
Supported Fieldbus specifications	PROFIBUS
Module model	TS5K-PB
Module type	Slave
Node address	The node address is specified by the user parameters. (1 to 125)
Communication speed	Auto-detect (9.6 kbps, 19.2 kbps, 45.45 kbps, 93.75 kbps, 187.5 kbps, 500 kbps, 1.5 Mbps, 3 Mbps, 6 Mbps, 12 Mbps)
GSD file	Each device over a PROFIBUS network is associated with an GSD file that contains all required information for the device. The GSD file can be used under the network configuration. This file can be loaded to make the basic settings. The GSD file is included in the Instruction Manual CD.
Number of inputs	512 inputs DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, DIN1001 to DIN1064
Number of outputs	512 outputs DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, DOUT1001 to DOUT1064

4.2 Hardware Configuration

4.2.1 LED Screen

Front View

#	Item
1	Operation mode LED
2	Status LED
3	PROFIBUS connector D-sub 9-pin



Operation mode LED

LED status	Description
Off	Offline/Power is turned off
Solid green	Online/Normal operation
Blinking green	Online/Clear
Blinking red (1 time)	Parameter error
Blinking red (2 times)	PROFIBUS configuration error

Status LED

LED status	Description
Off	Not initialized.
Green	Initialization complete.
Blinking green	Initialization complete. Diagnosing in progress (Extended diagnostic bit is set)
Red	Exception error

4.2.2 Connector Pin Arrangement

Pin	Signal name	Application
1	-	Not connected
2	-	Not connected
3	RXD (B-Line)	RxD/TxD RS485 level
4	RTS	Send request
5	GND (V-)	Bus ground (isolated)
6	P5V (V+)	+5 V termination power (isolated, short circuit protection)
7	-	Not connected
8	TXD (A-Line)	RxD/TxD RS485 level
9	-	Not connected
Housing	Shield	Internally connected to the Anybus protective ground via a cable shield filter Compliant with PROFIBUS standard

Field bus connector: 9-pin D_SUB (female)

4.2.3 Cables

For the cables used for the external connections, use the cables recommended for the master device that you use.
Model: XM2D-0901 or equivalent

4.3 Parameters

The PROFIBUS settings are configured using P012 in the PLC.PAR file.

For details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

[P012] PROFIBUS SETTING

Parameter key	PROFIBUS-NODE	
Parameter data	= (Node)	
Data type	Integer type	
Data count	1	
Data range	0 to 125	
Description	(Node)	This sets the Fieldbus node address. Set the node address so that it does not duplicate the address of any other devices on the same network. The range of setting values varies depending on the Fieldbus type.
Default	= 0	

Setting example:

Setting a node address of 2

```
‘[P012]    PROFIBUS SETTING
{ Node Addr}
{0-125 }
PROFIBUS-NODE = 2
```

5. DeviceNet Slave Module

5.1 Specifications

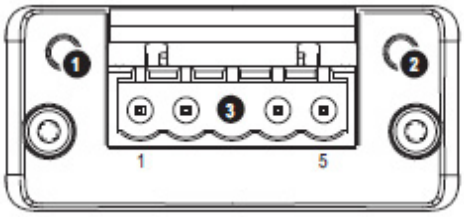
Supported Fieldbus specifications		DeviceNet
Module model		TS5K-DV
Module type		Slave
Scanner operation		Not supported
Node address		Each DeviceNet is assigned its own unique node address on the network. The node address is a value from 1 to 63 that is used to identify each node. It is specified by the PLC parameters.
Communication speed		It is specified by the PLC parameters. 0: 125 kbps 1: 250 kbps 2: 500 kbps
EDS file		Each device over a DeviceNet network is associated with an EDS file that contains all required information for the device. The EDS file can be used under the network configuration. This file can be loaded to make the basic settings. The EDS file is included in the Instruction Manual CD.
Explicit message		Not supported
I/O message		Not supported
Predefined Master/Slave ConnectionSet		
	Polling	Supported
	Bit Strobe	Supported
	Cyclic	Supported
	Change of State	Supported
Number of inputs		512 inputs DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, DIN1001 to DIN1064
Number of outputs		512 outputs DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, DOUT1001 to DOUT1064

5.2 Hardware Configuration

5.2.1 LED Screen

Front View

#	Item
1	Network LED
2	Module LED
3	DeviceNet connector



Network LED

LED status	Description
Off	Offline/Power is not being supplied to the network.
Solid green	Online, Normal operation
Blinking green	Online, Connection has been established.
Solid red	Critical link failure
Blinking red	Connection timed out.
Red/Green	Self-test

Module LED

LED status	Description
Off	No error was detected (power is turned off)
Solid green	Normal operation
Blinking green	Configuration is incorrect.
Solid red	Critical failure
Blinking red	Recoverable fault
Red/Green	Self-test

5.2.2 Connector Pin Arrangement

Pin	Signal name	Application
1	24G (V-)	Negative power supply voltage
2	CAN_L	CAN_L bus line
3	SHIELD	Shield cable
4	CAN_H	CAN_H bus line
5	P24V (V+)	Positive power supply voltage

Fieldbus connector: 5.08 pluggable screw

5.2.3 Cables

For the cables used for the external connections, use the cables recommended for the master device that you use.
Model: MSTB2.5/5 - STF-5.08 or equivalent

5.3 Parameters

The DeviceNet settings are configured using P013 in the PLC.PAR file.

For details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

[P013] DEVICENET SETTING

Parameter key	DEVICENET-NODE
Parameter data	= (Node)
Data type	Integer type
Data count	1
Data range	0 to 63
Description	This sets the Fieldbus node address. Set the node address so that it does not duplicate the address of any other devices on the same network. The range of setting values varies depending on the Fieldbus type.
	0: No Fieldbus options 1 to 63: Node address valid range in DeviceNet
Default	1

Parameter key	DEVICENET-SPEED
Parameter data	= (Communication speed)
Data type	Integer type
Data count	1
Data range	0 to 2
Description	This sets the communication speed. Set it to match the communication speed of the master device. No Fieldbus options (initial value) 0: 125 kbps 1: 250 kbps 2: 500 kbps
Default	0

Setting example:

Setting a node address of 1 and communication speed of 125 kbps

```
‘[P013]    DEVICENET SETTING
‘{Node Addr}
‘{0:DISABLE, 1-63: ENABLE}
DEVICENET-NODE = 1
‘{Speed}
DEVICENET-SPEED = 0
```

6. PROFINET Slave Module

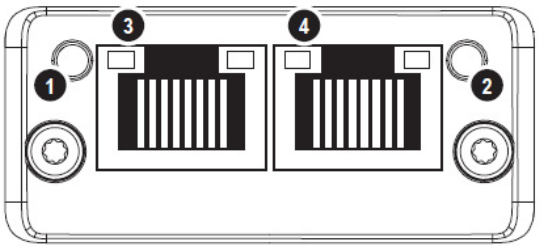
6.1 Specifications

Supported Fieldbus specifications	PROFINET
Module model	TS5K-PN
Module type	IO device
Supported master device	S7-1200 by Siemens AG
GSDML file	Each device over a PROFINET network is associated with an GSDML file that contains all required information for the device. The GSDML file can be used under the network configuration. This file can be loaded to make the basic settings. **The GSDML file is included in the Instruction Manual CD.** **Select a GSDML file according to the Fieldbus module version. For verifying the Fieldbus module version, refer to "*16 Verifying the Module Version*".**
	Fieldbus module version: M40 PN 1.53.1 **File name:** **"GSDML-V2.35-SHIBAURAMACHINE-TS5K-PN-20211001.xml"**
	Fieldbus module version: M40 PN 2.5.1 **File name:** **"GSDML-V2.41-SHIBAURA MACHINE CO.,LTD.-TS5K-PN-20230615.xml"**
Number of inputs	512 inputs DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, DIN1001 to DIN1064
Number of outputs	512 outputs DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, DOUT1001 to DOUT1064
Number of input data items	16 words D700 to D70F
Number of output data items	16 words D740 to D74F
Supported software version	Main unit: X8KCS-03D or later

6.2 Hardware Configuration

6.2.1 LED Screen

Front View

#	Item	Connector	
1	Network status LED	Ethernet, RJ45	
2	Module status LED		
3	Link/Activity LED (Port 1)		
4	Link/Activity LED (Port 2)		

Network status LED

LED status	Description
Off	Power off or no connection with controller
Solid green	Online, Connection with controller established (Run state)
Blinking green (1 time)	Online, Controller in stop state or IO data error, IRT synchronization is incomplete
Blinking green	Identifying node on network
Solid red	Critical error (internal error)
Blinking red (1 time)	Station name is not set
Blinking red (2 times)	IP address is not set
Blinking red (3 times)	Setting does not match actual identified value

Module status LED

LED description	Description
Off	Power is not turned on (Setup in progress or NW_INIT state)
Solid green	Normal state
Blinking green	Diagnosing event
Solid red	Exception error (exception state, critical error, etc.)
Blinking red	Recoverable fault (do not turn off the power)

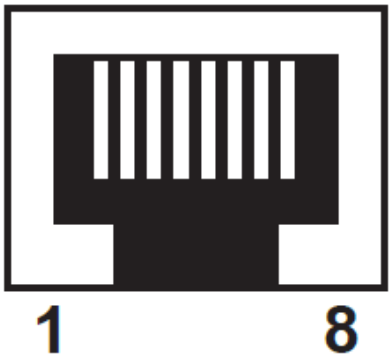
Link/Activity LED 3/4

LED status	Description
Off	No link, No communication
Solid green	Link established, No communication
Blinking green	Link established, Communicating

6.2.2 Connector Pin Arrangement

Same for Port 1 and Port 2

Pin	Signal name
1	TXD+
2	TXD-
3	RXD+
4	-
5	-
6	RXD-
7	-
8	-



6.2.3 Cables

For the cables used for the external connections, use the cables recommended for the master device that you use.

RJ45 connector: Cat 5 shielded cable (straight cable)

6.3 Parameters

The PROFINET settings are configured using P014 in the PLC.PAR file.

For details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

[P014] PROFINET SETTING

Parameter key	PROFINET-DCP-ENABLE
Parameter data	= (IP address specified by DCP)
Data type	Integer type
Data count	1
Data range	0 to 1
Description	0: Disable specifying of IP address by DCP (The IP address that was set in the PLC.PAR file is enabled.) 1: Enable specifying of IP address by DCP (The IP address that was set in the PLC.PAR file is disabled.)
Default	= 1

Parameter key	PROFINET-ADD
Parameter data	= (IP address)
Data type	Integer type
Data count	4
Data range	-
Description	- The IP address is an address for identifying the EtherNet connection device. - The address is expressed using four numbers from 0 to 255 separated by a comma (,). - If a PC connected to the PROFINET and network is connected directly using a 1-to-1 connection, the initial value (any selected) can be used for the IP address. However, if the PC is connected to an existing LAN, set the IP address based on instructions from your LAN system administrator. If IP addresses are duplicated, the functions may not work correctly. - The PC or PLC that communicates with the PROFINET must be connected to the same network.
Default	= 192,168,0,1

Parameter key	PROFINET
Parameter data	= (Subnet mask)
Data type	Integer type
Data count	4
Data range	-
Description	This specifies the subnet mask of the connected network. This must be set when connecting to an existing LAN. Set the subnet mask based on instructions from your LAN system administrator.
Default	=255,255,255,0

Parameter key	PROFINET -GATEWAY
Parameter data	= (Gateway)
Data type	Integer type
Data count	4
Data range	-
Description	This specifies the default gateway address. This must be set when connecting to an existing LAN. Set the default gateway address based on instructions from your LAN system administrator.
Default	=0.0,0.0

Setting example:

When PROFINET is selected and the following is set in the S7-1200 made by Seimens AG.

Enable IP address by DCP

IP address: 192.168.0.10

Subnet mask: 255.255.255.0

Gateway: 0.0.0.0

```
'[P014] PROFINET SETTING
'{DCP Enable}
PROFINET-DCP-ENABLE = 1
'{IPAdress}
PROFINET-ADD = 192, 168, 0, 10
'{Net Mask}
PROFINET-MASK = 255, 255, 255, 0
'{GateWay}
PROFINET-GATEWAY = 0, 0, 0, 0
```

7. EtherCAT Slave Module

7.1 Specifications

Supported Fieldbus specifications	EtherCAT
Module model	TS5K-CA
Module type	IO device
Supported master device	NJ-301 by Omron
ESI file	Each device over an EtherCAT network is associated with an ESI file that contains all required information for the device. The ESI file can be used under the network configuration. This file can be loaded to make the basic settings. **The ESI file is included in the Instruction Manual CD.** File name: Shibaura Machine TS5K-CA V1.0.1.xml
Number of inputs	512 inputs DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, DIN1001 to DIN1064
Number of outputs	512 outputs DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, DOUT1001 to DOUT1064
Number of input data items	16 words D700 to D70F
Number of output data items	16 words D740 to D74F
Supported software version	Main unit: X8KCS-03D or later
Remarks	EoE is not supported. The ID settings support Station Alias (specifying of node address from EtherCAT master device).

* These specifications are for an EtherCAT compliant series.

Registered trademarks

EtherCAT® is a registered trademark and patented technology licensed by Beckhoff Automation GmbH of Germany.

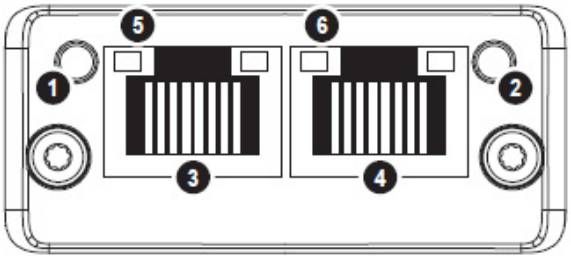
Reference documentation

- ETG.1000.5 EtherCAT Specification, Part 5 - Application layer service definition
- ETG.1000.6 EtherCAT Specification, Part 6 - Application layer protocol specification
- ETG.1300 Indicator and Labeling Specification
- ETG.6010 Implementation Directive for CiA402
- ETG.9001 EtherCAT Marking Rules

7.2 Hardware Configuration

7.2.1 LED Screen

Front View

#	Item	Connector	
1	RUN LED	Ethernet, RJ45	
2	ERROR LED		
3	EtherCAT (IN port)		
4	EtherCAT (OUT port)		
5	Link/Activity (IN port)		
6	Link/Activity (OUT port)		

RUN LED

LED status	Description
Off	"INIT" state (or power off)
Solid green	"OPERATIONAL" state
Blinking green (1 time)	"SAFE-OPERATIONAL" state
Blinking green	"PRE-OPERATIONAL" state
Solid red	When both the RUN LED and ERR LED are solid red, this indicates that a critical error has occurred. The bus interface is in a physically passive state.

ERR LED

LED status	Description
Off	No error occurred (or power off)
Blinking red	Cannot change to the state that was sent by the master device because the register or object setting is invalid.
Blinking red (1 time)	Applications on slave devices autonomously change the state of EtherCAT.
Blinking red (2 times)	Sink manager watchdog timed out (default time interval is 100 ms).
Solid red	EXCEPTION state. When both the RUN LED and ERR LED are solid red, this indicates that a critical error has occurred. The bus interface is in a physically passive state.
Rapid blinking	An error occurred during the boot process.

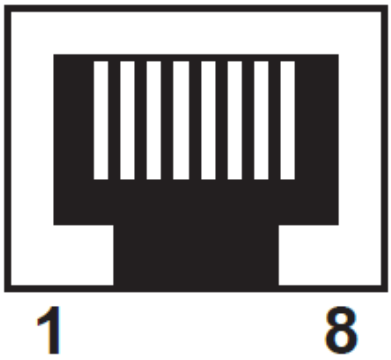
Link/Activity LED

LED status	Description
Off	No link, No communication (or power off)
Solid green	Link established, No communication
Blinking green	Link established, Communicating

7.2.2 Connector Pin Arrangement

Same for Port 1 and Port 2

Pin	Signal name
1	TXD+
2	TXD-
3	RXD+
4	-
5	-
6	RXD-
7	-
8	-



7.2.3 Cables

For the cables used for the external connections, use the cables recommended for the master device that you use.

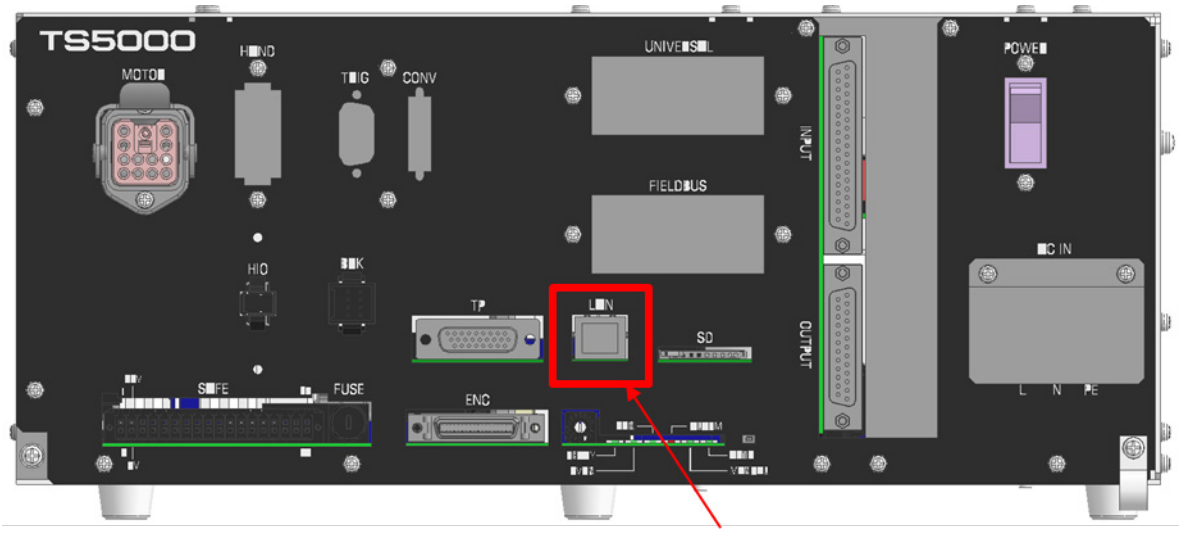
RJ45 connector: Cat 5 shielded cable (straight cable)

8. Modbus TCP Server

8.1 Specifications

Supported Fieldbus specification	Modbus TCP Server
IP address	Determined according to the IP-ADDRESS parameter in the ETHERNET.PAR file
Communication port	502
Number of inputs	512 inputs DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, DIN1001 to DIN1064
Number of outputs	512 outputs DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, DOUT1001 to DOUT1064
Number of input data items	16 words D700 to D70F
Number of output data items	16 words D740 to D74F
Supported software version	Main unit: X8KCS-05D or later
Remarks	**This function cannot be used concurrently with any other Fieldbus options.** The Modbus TCP Server can be used as a Fieldbus only when the slave module for the Fieldbus is not used. To use this function, use the Ethernet connector that is attached to the controller as the standard.

8.2 Hardware Configuration



Ethernet connector

8.2.1 Cable

RJ45 connector: Cat5 shielded cable (straight cable)

8.3 Parameters

Use [P016] in the PLC.PAR file for setting the Modbus.

For the details on the parameter setting procedure, see the *STA1166 Parameter Manual*.

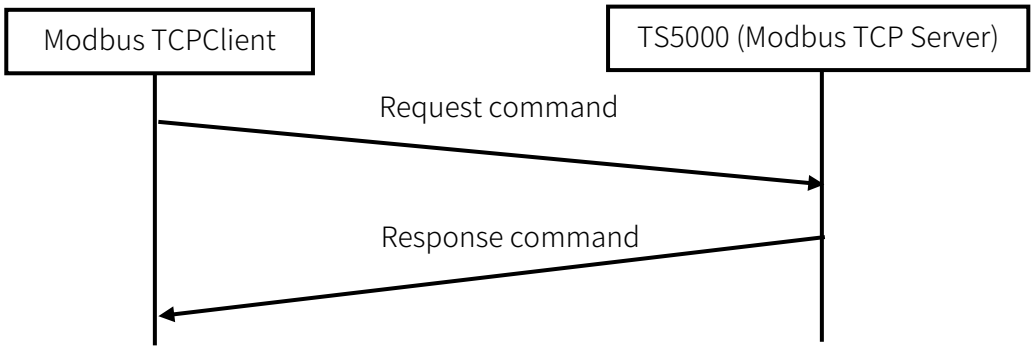
[P016] MODBUS PROTCOL SELECT

Parameter key	MODOBUS-KIND
Parameter data	= (Modbus protocol type)
Data type	Integer type
Data count	1
Data range	0 and 3
Description	0：Modbus function disabled 3：Modbus TCP Server function enabled
Default	= 0

8.4 Communication Specification

8.4.1 Communication Protocol

The Modbus TCP Server that is installed in TS5000 responds to a request command of the Modbus TCP Client.



8.4.2 Function codes

The table below lists the function codes that are supported. When an unsupported function code is received, the Modbus TP Server returns an error message.

Supported function code	Function	Maximum data size	Register range
0x01 (1)	Reads multiple bits from the register	128 bits	X000 to YF7F
0x02 (2)	Same operation as that of function code 0x01	128 bits	X000 to YF7F
0x03 (3)	Reads multiple words from the register	60 words	D000 to D77F
0x04 (4)	Same operation as that of function code 0x03	60 words	D000 to D77F
0x05 (5)	Writes single bit to the register	1 bit	X000 to YF7F
0x06 (6)	Writes single word to the register	1 word	D000 to D77F
0x0F (15)	Writes multiple bits to the register	128 bits	X000 to YF7F
0x10 (16)	Writes multiple words to the register	30 words	D000 to D77F

8.4.3 Addresses

The following table lists the registers corresponding to the addresses of the Modbus TCP of the Modbus/TCP Server function.

Register	Communication address
D000	0x0000
D001	0x0001
:	:
D07F	0x007F
D100	0x0080
D101	0x0081
:	:
D17F	0x00FF
D200	0x0100
:	:
D770	0x0380
:	:
D77F	0x03FF

Register	Communication address
X000	0x1000
X001	0x1001
:	:
X07F	0x107F
Y100	0x1080
Y101	0x1081
:	:
Y17F	0x10FF
X200	0x1100
:	:
YF00	0x17F0
:	:
YF7F	0x17FF

- Data can be written only to X2**, X4**, X6**, and X8** of the Fieldbus area of the XY register.
- Write operation towards an inaccessible area cannot be performed. In this case, 0 is returned for the read data.

8.4.4 Communication format of Each Function Code

This section describes the procedure for using each function code. Numerical values such as 00 and FF are handled as hexadecimal numbers.

0x01, 0x02

Function

Reads multiple bits from the registers.

Request command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x01 or 0x02
Data(Address)	2	Read start address
Data(Length)	2	Read bit length

Response command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x01 or 0x02
Data(Length)	2	Number of bytes read
Data	Variable length	The read data comprising the number of bytes that is specified in Data (Length) is stored.

Example of transmission/reception

In the following example, 32-bit data is read starting from the X110 register.

Register image

	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	HEX
Y11W	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0x1234
Y12W	0	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0x5678

Data transmitted: 00 00 00 00 00 06 00 01 10 90 00 20

Data received: 00 00 00 00 00 07 00 01 04 34 12 78 56

0x03, 0x04

Function

Reads multiple words from the registers.

Request command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x03 or 0x04
Data(Address)	2	Read start address
Data(Length)	2	Number of words read

Response command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x03 or 0x04
Data(Length)	1	Number of bytes read
Data	Variable length	The read data comprising the number of bytes that is specified in Data (Length) is stored.

Example of transmission/reception

In the following example, 2-word data is read starting from the D001 register.

Register image

	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	HEX
D001	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0x0002
D002	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0x0004

Data transmitted: 00 00 00 00 00 06 00 03 00 01 00 02

Data received: 00 00 00 00 00 07 00 03 04 00 02 00 04

0x05

Function

Writes a single bit to the register.

Request command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x05
Data(Address)	2	Write start address
Data(Length)	2	ON information ON：0xFF, 0x00 OFF：0x00, 0x00

Response command

A response is returned in the same format as that of the request format.

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x05
Data(Address)	2	Write start address
Data(Length)	2	ON/OFF information ON: 0xFF, 0x00 OFF: 0x0, 0x00

Example of transmission/reception

In the following example, ON is written to the X200 register.

Data transmitted: 00 00 00 00 00 06 00 05 11 00 FF 00

Data received: 00 00 00 00 00 06 00 05 11 00 FF 00

Register image

	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	HEX
X20W	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0x0001

0x06

Function

Writes a single word to the register.

Request command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x06
Data(Address)	2	Write start address
Data(Length)	2	Write data

Response command

A response is returned in the same format as that of the request format.

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x06
Data(Address)	2	Write start address
Data(Length)	2	Write data

Example of transmission/reception

In the following example, 0x5678 is written to the D001 register.

Register image

	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	HEX
D001	0	1	0	1	0	1	1	0	0	1	1	1	1	0	0	0	0x5678

Data transmitted: 00 00 00 00 00 06 00 06 00 01 56 78

Data received: 00 00 00 00 00 06 00 06 00 01 56 78

0x0F

Function

Writes multiple bits to the registers.

Request command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x0F
Data(Address)	2	Write start address
Data(Length)	2	Number of bits written
	1	Number of bytes written
	2	Write bit data

Response command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x0F
Data(Address)	2	Write start address
Data(Length)	2	Number of bits written

Example of transmission/reception

In the following example, 16-bit data of 0x1234 is written to the registers starting from the X200 register.

Data transmitted: 00 00 00 00 00 09 00 0F 11 00 00 10 02 34 12

Data received: 00 00 00 00 00 06 00 0F 11 00 00 10

Register image

	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	HEX
X20W	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0x1234

0x10

Function

Writes multiple words to the registers.

Request command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x10
Data(Address)	2	Write start address
Data(Length)	2	Number of words written
	2	Number of bytes written
	Variable length	The written data comprising the number of bytes that is specified in Data (Length) is stored.

Response command

Name	Size (Bye)	Contents
Transaction ID	2	0x00, 0x00
Protocol ID	2	0x00, 0x00
Length	2	Number of bytes following Unit ID
Unit ID	1	0x00
Function code	1	0x10
Data(Address)	2	Write start address
Data(Length)	2	Number of words written

Example of transmission/reception

In the following example, 2-word data of 0x1234 and 0x9ABC is written to the registers starting from the D001 register.

Data transmitted: 00 00 00 00 00 0B 00 10 00 01 00 02 04 12 34 9A BC

Data received: 00 00 00 00 00 06 00 10 00 01 00 02

Register image

	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	HEX
D001	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0x1234
D002	1	0	0	1	1	0	1	0	1	0	1	1	1	1	0	0	0x9ABC

9. Alarm Detection

The following alarms are detected by the Fieldbus Slave function.

- 8-353 Fieldbus Parameter error

The setting values for the Fieldbus parameters are different.
Set the parameters again, and restart the controller.

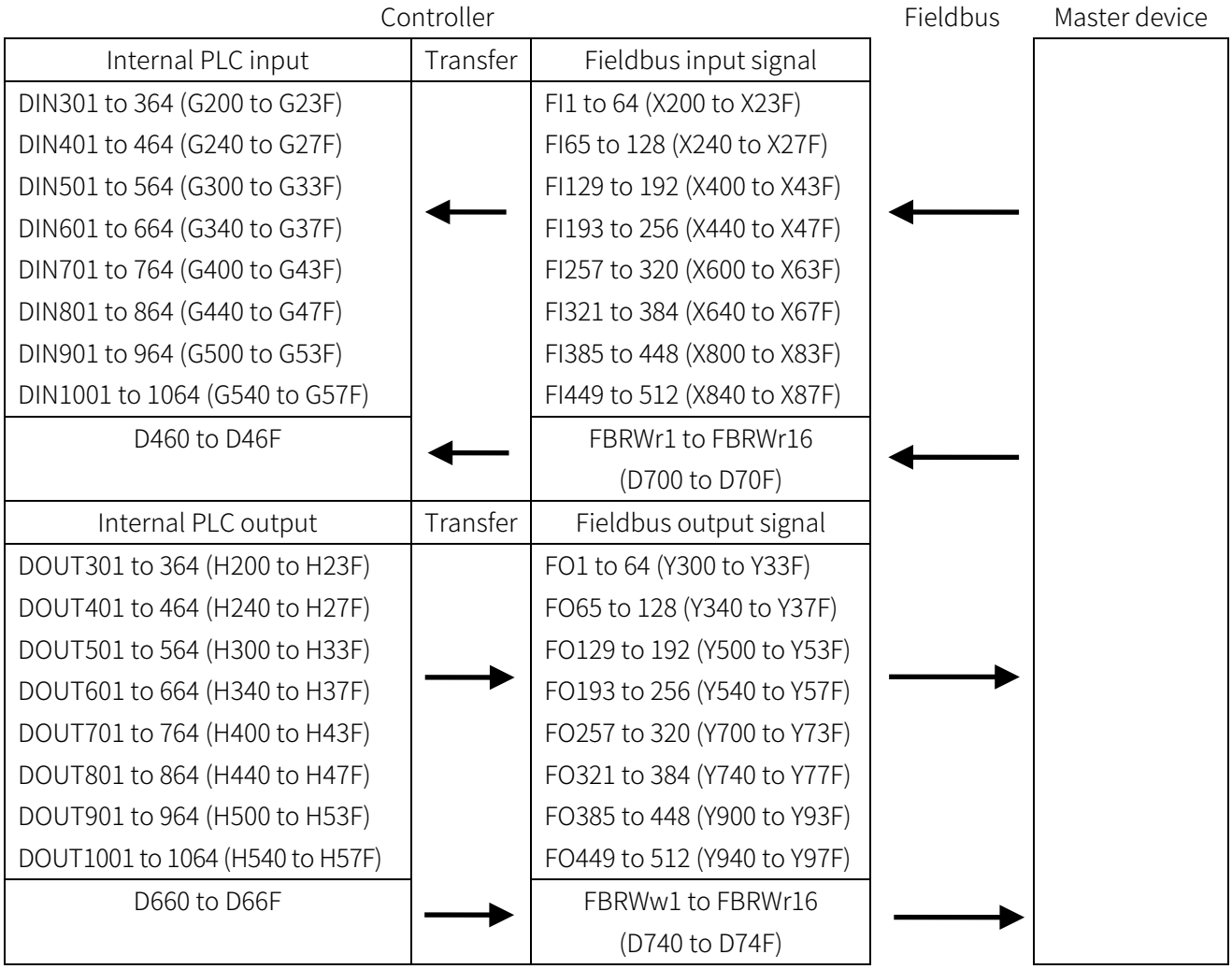
- 8-354 Fieldbus Offline

The controller and master device side are offline.
This alarm continues until the status changes to online.
If the robot is operating, an emergency stop is applied.
Find the specific cause of the offline status by checking at the Fieldbus master device.

- 8-355 Fieldbus Board error

A Fieldbus board error was detected.
The board may need to be replaced, and so contact the Shibaura Machine service center.

10. Input/Output Addresses



*1 The input/output addresses of the master device vary depending on the manufacturer of the connected PLC.

*2 The available bits and the data range vary depending on the Fieldbus type and settings.

11. TCmini (Internal PLC)

The controller includes TCmini (internal PLC). TCmini (internal PLC) can be used to input and output any selected input/output signals to the Fieldbus.

This section provides address tables for Fieldbus and the TCmini that performs input and output only. For details on TCmini (internal PLC), see the "*STA1168_TCmini (Internal PLC) Manual*".

11.1 Bit Input (FieldBUS→TCmini)

Address Table

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X20W	FI16	FI15	FI14	FI13	FI12	FI11	FI10	FI9	FI8	FI7	FI6	FI5	FI4	FI3	FI2	FI1
X21W	FI32	FI31	FI30	FI29	FI28	FI27	FI26	FI25	FI24	FI23	FI22	FI21	FI20	FI19	FI18	FI17
X22W	FI48	FI47	FI46	FI45	FI44	FI43	FI42	FI41	FI40	FI39	FI38	FI37	FI36	FI35	FI34	FI33
X23W	FI64	FI63	FI62	FI61	FI60	FI59	FI58	EI57	FI56	FI55	FI54	FI53	FI52	FI51	FI50	FI49
X24W	FI80	FI79	FI78	FI77	FI76	FI75	FI74	FI73	FI72	FI71	FI70	FI69	FI68	FI67	FI66	FI65
X25W	FI96	FI95	FI94	FI93	FI92	FI91	FI90	FI89	FI88	FI87	FI86	FI85	FI84	FI83	FI82	FI81
X26W	FI112	FI111	FI110	FI109	FI108	FI107	FI106	FI105	FI104	FI103	FI102	FI101	FI100	FI99	FI98	FI97
X27W	FI128	FI127	FI126	FI125	FI124	FI123	FI122	FI121	FI120	FI119	FI118	FI117	FI116	FI115	FI114	FI113
X40W	FI144	FI143	FI142	FI141	FI140	FI139	FI138	FI137	FI136	FI135	FI134	FI133	FI132	FI131	FI130	FI129
X41W	FI160	FI159	FI158	FI157	FI156	FI155	FI154	FI153	FI152	FI151	FI150	FI149	FI148	FI147	FI146	FI145
X42W	FI176	FI175	FI174	FI173	FI172	FI171	FI170	FI169	FI168	FI167	FI166	FI165	FI164	FI163	FI162	FI161
X43W	FI192	FI191	FI190	FI189	FI188	FI187	FI186	FI185	FI184	FI183	FI182	FI181	FI180	FI179	FI178	FI177
X44W	FI208	FI207	FI206	FI205	FI204	FI203	FI202	FI201	FI200	FI199	FI198	FI197	FI196	FI195	FI194	FI193
X45W	FI224	FI223	FI222	FI221	FI220	FI219	FI218	FI217	FI216	FI215	FI214	FI213	FI212	FI211	FI210	FI209
X46W	FI240	FI239	FI238	FI237	FI236	FI235	FI234	FI233	FI232	FI231	FI230	FI229	FI228	FI227	FI226	FI225
X47W	FI256	FI255	FI254	FI253	FI252	FI251	FI250	FI249	FI248	FI247	FI246	FI245	FI244	FI243	FI242	FI241
X60W	FI272	FI271	FI270	FI269	FI268	FI267	FI266	FI265	FI264	FI263	FI262	FI261	FI260	FI259	FI258	FI257
X61W	FI288	FI287	FI286	FI285	FI284	FI283	FI282	FI281	FI280	FI279	FI278	FI277	FI276	FI275	FI274	FI273
X62W	FI304	FI303	FI302	FI301	FI300	FI299	FI298	FI297	FI296	FI295	FI294	FI293	FI292	FI291	FI290	FI289
X63W	FI320	FI319	FI318	FI317	FI316	FI315	FI314	FI313	FI312	FI311	FI310	FI309	FI308	FI307	FI306	FI305
X64W	FI336	FI335	FI334	FI333	FI332	FI331	FI330	FI329	FI328	FI327	FI326	FI325	FI324	FI323	FI322	FI321
X65W	FI352	FI351	FI350	FI349	FI348	FI347	FI346	FI345	FI344	FI343	FI342	FI341	FI340	FI339	FI338	FI337
X66W	FI368	FI367	FI366	FI365	FI364	FI363	FI362	FI361	FI360	FI359	FI358	FI357	FI356	FI355	FI354	FI353
X67W	FI384	FI383	FI382	FI381	FI380	FI379	FI378	FI377	FI376	FI375	FI374	FI373	FI372	FI371	FI370	FI369
X80W	FI400	FI399	FI398	FI397	FI396	FI395	FI394	FI393	FI392	FI391	FI390	FI389	FI388	FI387	FI386	FI385
X81W	FI416	FI415	FI414	FI413	FI412	FI411	FI410	FI409	FI408	FI407	FI406	FI405	FI404	FI403	FI402	FI401
X82W	FI432	FI431	FI430	FI429	FI428	FI427	FI426	FI425	FI424	FI423	FI422	FI421	FI420	FI419	FI418	FI417
X83W	FI448	FI447	FI446	FI445	FI444	FI443	FI442	FI441	FI440	FI439	FI438	FI437	FI436	FI435	FI434	FI433
X84W	FI464	FI463	FI462	FI461	FI460	FI459	FI458	FI457	FI456	FI455	FI454	FI453	FI452	FI451	FI450	FI449
X85W	FI480	FI479	FI478	FI477	FI476	FI475	FI474	FI473	FI472	FI471	FI470	FI469	FI468	FI467	FI466	FI465
X86W	FI496	FI495	FI494	FI493	FI492	FI491	FI490	FI489	FI488	FI487	FI486	FI485	FI484	FI483	F482I	FI481
X87W	FI512	FI511	FI510	FI509	FI508	FI507	FI506	FI505	FI504	FI503	FI502	FI501	FI500	FI499	FI498	FI497

11.2 Bit Output (TCmini→FieldBUS)

Address Table

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Y30W	FO16	FO15	FO14	FO13	FO12	FO11	FO10	FO9	FO8	FO7	FO6	FO5	FO4	FO3	FO2	FO1
Y31W	FO32	FO31	FO30	FO29	FO28	FO27	FO26	FO25	FO24	FO23	FO22	FO21	FO20	FO19	FO18	FO17
Y32W	FO48	FO47	FO46	FO45	FO44	FO43	FO42	FO41	FO40	FO39	FO38	FO37	FO36	FO35	FO34	FO33
Y33W	FO64	FO63	FO62	FO61	FO60	FO59	FO58	EO57	FO56	FO55	FO54	FO53	FO52	FO51	FO50	FO49
Y34W	FO80	FO79	FO78	FO77	FO76	FO75	FO74	FO73	FO72	FO71	FO70	FO69	FO68	FO67	FO66	FO65
Y35W	FO96	FO95	FO94	FO93	FO92	FO91	FO90	FO89	FO88	FO87	FO86	FO85	FO84	FO83	FO82	FO81
Y36W	FO112	FO111	FO110	FO109	FO108	FO107	FO106	FO105	FO104	FO103	FO102	FO101	FO100	FO99	FO98	FO97
Y37W	FO128	FO127	FO126	FO125	FO124	FO123	FO122	FO121	FO120	FO119	FO118	FO117	FO116	FO115	FO114	FO113
Y50W	FO144	FO143	FO142	FO141	FO140	FO139	FO138	FO137	FO136	FO135	FO134	FO133	FO132	FO131	FO130	FO129
Y51W	FO160	FO159	FO158	FO157	FO156	FO155	FO154	FO153	FO152	FO151	FO150	FO149	FO148	FO147	FO146	FO145
Y52W	FO176	FO175	FO174	FO173	FO172	FO171	FO170	FO169	FO168	FO167	FO166	FO165	FO164	FO163	FO162	FO161
Y53W	FO192	FO191	FO190	FO189	FO188	FO187	FO186	FO185	FO184	FO183	FO182	FO181	FO180	FO179	FO178	FO177
Y54W	FO208	FO207	FO206	FO205	FO204	FO203	FO202	FO201	FO200	FO199	FO198	FO197	FO196	FO195	FO194	FO193
Y55W	FO224	FO223	FO222	FO221	FO220	FO219	FO218	FO217	FO216	FO215	FO214	FO213	FO212	FO211	FO210	FO209
Y56W	FO240	FO239	FO238	FO237	FO236	FO235	FO234	FO233	FO232	FO231	FO230	FO229	FO228	FO227	FO226	FO225
Y57W	FO256	FO255	FO254	FO253	FO252	FO251	FO250	FO249	FO248	FO247	FO246	FO245	FO244	FO243	FO242	FO241
Y70W	FO272	FO271	FO270	FO269	FO268	FO267	FO266	FO265	FO264	FO263	FO262	FO261	FO260	FO259	FO258	FO257
Y71W	FO288	FO287	FO286	FO285	FO284	FO283	FO282	FO281	FO280	FO279	FO278	FO277	FO276	FO275	FO274	FO273
Y72W	FO304	FO303	FO302	FO301	FO300	FO299	FO298	FO297	FO296	FO295	FO294	FO293	FO292	FO291	FO290	FO289
Y73W	FO320	FO319	FO318	FO317	FO316	FO315	FO314	FO313	FO312	FO311	FO310	FO309	FO308	FO307	FO306	FO305
Y74W	FO336	FO335	FO334	FO333	FO332	FO331	FO330	FO329	FO328	FO327	FO326	FO325	FO324	FO323	FO322	FO321
Y75W	FO352	FO351	FO350	FO349	FO348	FO347	FO346	FO345	FO344	FO343	FO342	FO341	FO340	FO339	FO338	FO337
Y76W	FO368	FO367	FO366	FO365	FO364	FO363	FO362	FO361	FO360	FO359	FO358	FO357	FO356	FO355	FO354	FO353
Y77W	FO384	FO383	FO382	FO381	FO380	FO379	FO378	FO377	FO376	FO375	FO374	FO373	FO372	FO371	FO370	FO369
Y90W	FO400	FO399	FO398	FO397	FO396	FO395	FO394	FO393	FO392	FO391	FO390	FO389	FO388	FO387	FO386	FO385
Y91W	FO416	FO415	FO414	FO413	FO412	FO411	FO410	FO409	FO408	FO407	FO406	FO405	FO404	FO403	FO402	FO401
Y92W	FO432	FO431	FO430	FO429	FO428	FO427	FO426	FO425	FO424	FO423	FO422	FO421	FO420	FO419	FO418	FO417
Y93W	FO448	FO447	FO446	FO445	FO444	FO443	FO442	FO441	FO440	FO439	FO438	FO437	FO436	FO435	FO434	FO433
Y94W	FO464	FO463	FO462	FO461	FO460	FO459	FO458	FO457	FO456	FO455	FO454	FO453	FO452	FO451	FO450	FO449
Y95W	FO480	FO479	FO478	FO477	FO476	FO475	FO474	FO473	FO472	FO471	FO470	FO469	FO468	FO467	FO466	FO465
Y96W	FO496	FO495	FO494	FO493	FO492	FO491	FO490	FO489	FO488	FO487	FO486	FO485	FO484	FO483	F482O	FO481
Y97W	FO512	FO511	FO510	FO509	FO508	FO507	FO506	FO505	FO504	FO503	FO502	FO501	FO500	FO499	FO498	FO497

11.3 Data Input (FieldBUS→TCmini)

This function is available in CC-Link, EtherNET/IP, PROFINET, and EtherCAT only.

Address Table

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D70*	FB RWr16	FB RWr15	FB RWr14	FB RWr13	FB RWr12	FB RWr11	FB RWr10	FB RWr9	FB RWr8	FB RWr7	FB RWr6	FB RWr5	FB RWr4	FB RWr3	FB RWr2	FB RWr1
D71*	FB RWr32	FB RWr31	FB RWr30	FB RWr29	FB RWr28	FB RWr27	FB RWr26	FB RWr25	FB RWr24	FB RWr23	FB RWr22	FB RWr21	FB RWr20	FB RWr19	FB RWr18	FB RWr17

D71* can be used only when 4 is set for the number of stations used and 2 is set for the expansion cyclic settings in CC-LINK.

11.4 Data Output (TCmini→FieldBUS)

This function is available in CC-Link, EtherNET/IP, PROFINET, and EtherCAT only.

Address Table

Register	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
D74*	FB RWw16	FB RWw15	FB RWw14	FB RWw13	FB RWw12	FB RWw11	FB RWw10	FB RWw9	FB RWw8	FB RWw7	FB RWw6	FB RWw5	FB RWw4	FB RWw3	FB RWw2	FB RWw1
D75*	FB RWw32	FB RWw31	FB RWw30	FB RWw29	FB RWw28	FB RWw27	FB RWw26	FB RWw25	FB RWw24	FB RWw23	FB RWw22	FB RWw21	FB RWw20	FB RWw19	FB RWw18	FB RWw17

D75* can be used only when 4 is set for the number of stations used and 2 is set for the expansion cyclic settings in CC-LINK.

Operation

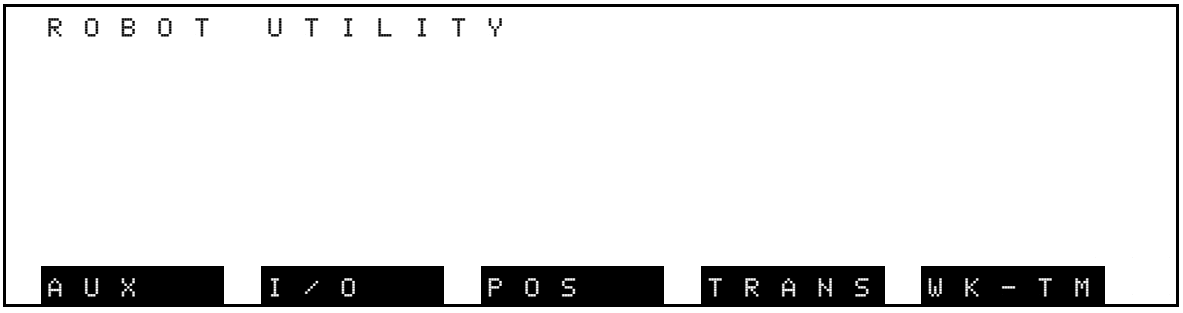
This section describes the operating procedures for the Fieldbus slave of the controller.

12. Operation Verification

The Fieldbus input signals are assigned to DIN301 to DIN364, DIN401 to DIN464, DIN501 to DIN564, DIN601 to DIN664, DIN701 to DIN764, DIN801 to DIN864, DIN901 to DIN964, and DIN1001 to DIN1064, and the output signals are assigned to DOUT301 to DOUT364, DOUT401 to DOUT464, DOUT501 to DOUT564, DOUT601 to DOUT664, DOUT701 to DOUT764, DOUT801 to DOUT864, DOUT901 to DOUT964, and DOUT1001 to DOUT1064. (The available bits and the data range vary depending on the Fieldbus type and settings.) Operation of the above signal lines can be verified in Utility mode. For the detailed operations in Utility mode, see the "Operation Manual". The procedure is shown below. (Check that the user parameter settings and settings by the Fieldbus master device were completed and that no alarm has occurred in the Fieldbus or controller.)

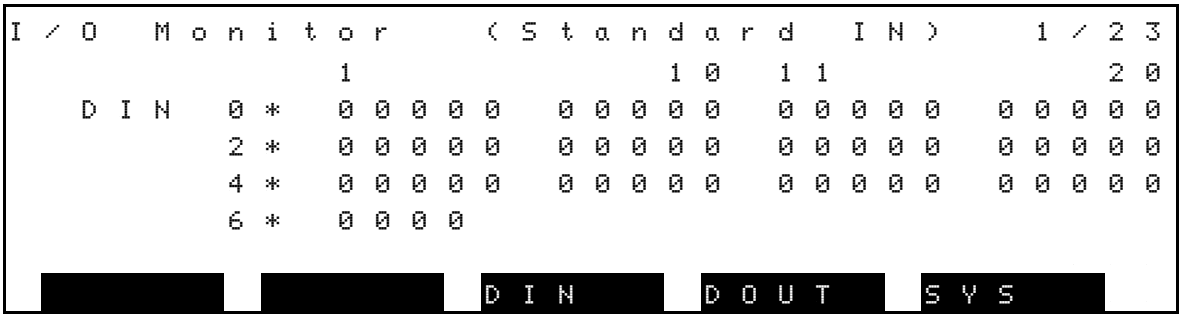
1 Press the teach pendant UTILITY key.

This switches to Utility mode, and the following display screen appears.



2 Pressing the teach pendant F2 key and I/O will switch to the external input signal display screen.

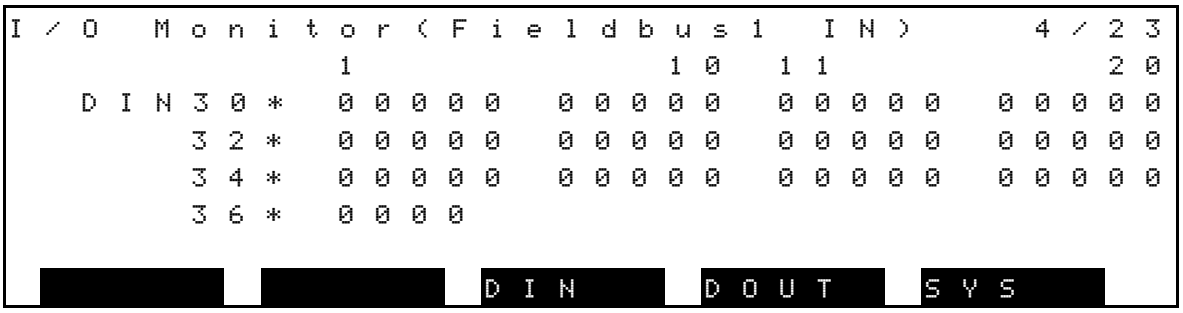
A screen displaying the status of general-purpose input signals appears as shown below.



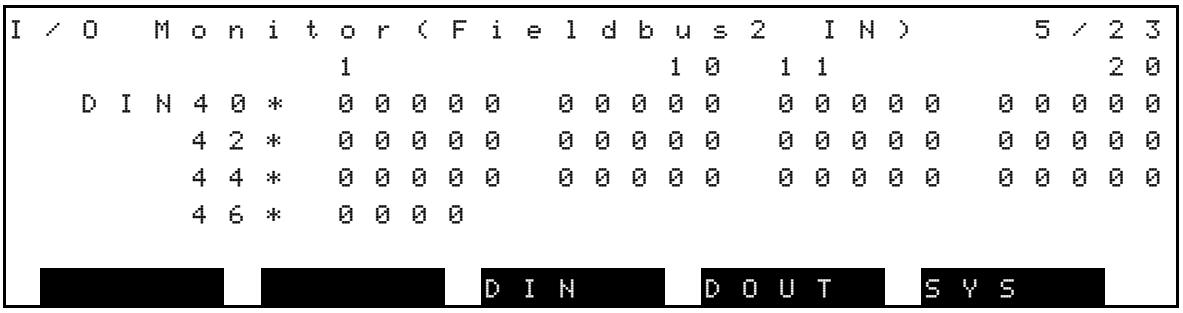
3 Press NEXT or ALT + ↓ three times.

The Fieldbus input is displayed. The 512 Fieldbus input signals assigned by master device operation can be verified as ON or OFF. The status of the input/output signal is "1" when the contact is ON (closed status) and is "0" when the contact is OFF (open status). The display shows 20 signals per row, divided into sections of 5 signals each, and the first signal number appears at the left edge.

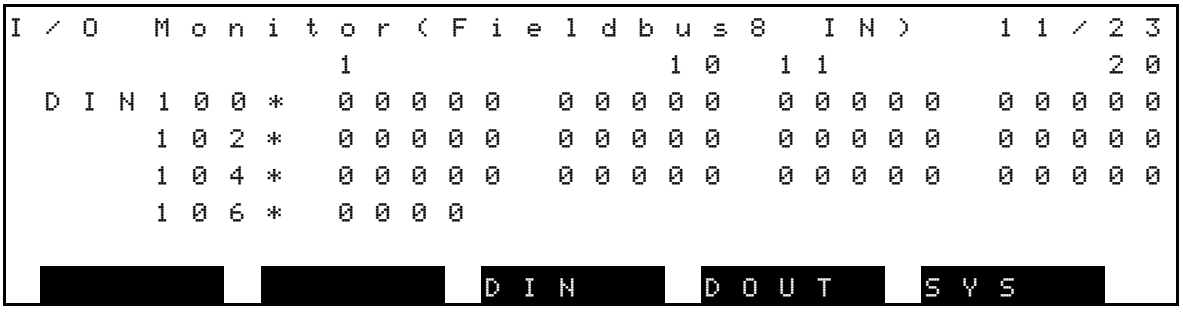
(Fieldbus input) 1st page



(Fieldbus input) 2nd page



(Fieldbus input) 8th page



4 Press DOUT to open the output screen.

Next, press NEXT or ALT + ↓ three times. The Fieldbus output is displayed. The 512 Fieldbus output signals can be verified as ON or OFF. The valid keys and their actions are shown in the following table.

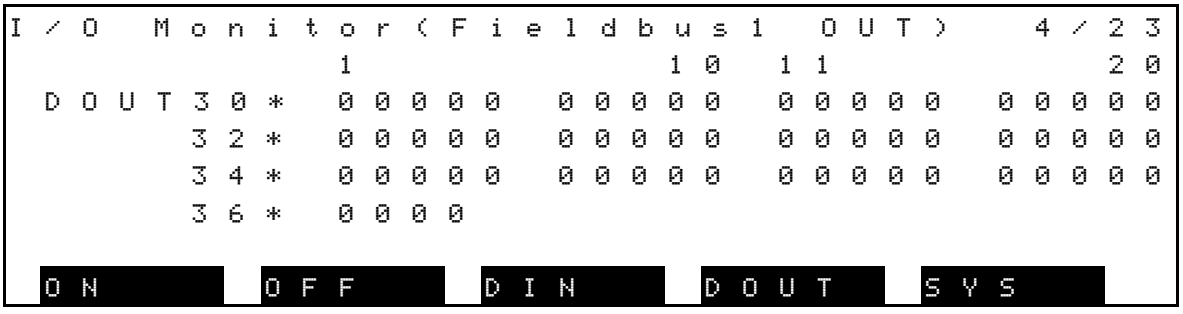
Key	Action description
←	This moves the cursor to the left. The cursor does not move if it is at the left edge.
→	This moves the cursor to the right. The cursor does not move if it is at the right edge.
↑	This moves the cursor up. The cursor does not move if it is at the top edge.
↓	This moves the cursor down. The cursor does not move if it is at the bottom edge.
ALT + ↓	This switches to the next page. This does not switch the page if it is already on the last page.
ALT + ↑	This switches to the previous page. This does not switch the page if it is already on the first page.
ESC	This returns to the utility screen.

The soft keys and their actions are shown in the following table.

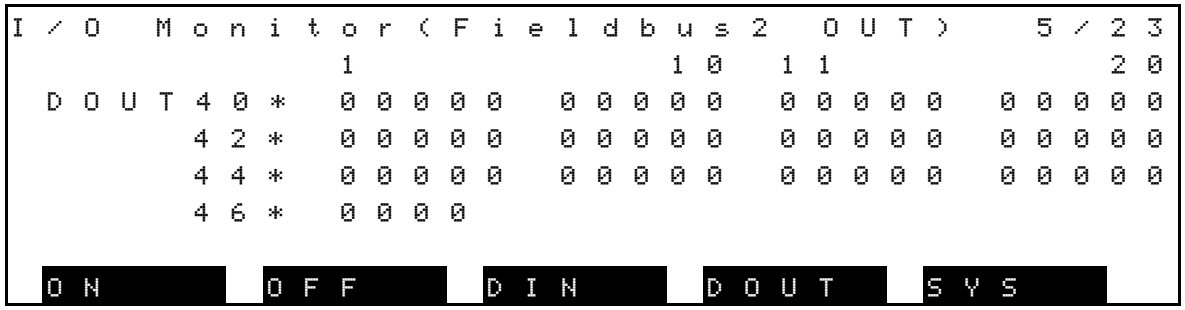
Soft key	Action description
[ON]	This turns on the bit where the cursor is located.
[OFF]	This turns off the bit where the cursor is located.

Operation of the signals assigned by the master device can be verified as ON or OFF.

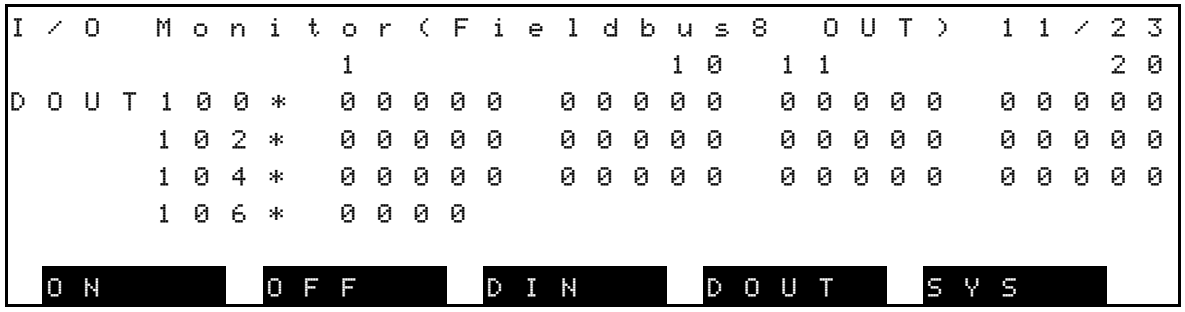
(Fieldbus output) 1st page



(Fieldbus output) 2nd page



(Fieldbus output) 8th page



13. Fieldbus System Signal Input/Output Function

This function assigns system signal input/output to the Fieldbus.
The value for the PLC parameter [P003] SEQUENCE FUNCTION SELECT SWITCH can be set as shown below so that the Fieldbus assignments for the Fieldbus input address [X26W] or [X27W] and the Fieldbus output address [Y36W] or [Y37W] are switched from general-purpose input/output to system input/output signals.

0: Disable the function (Assigned from main system connector)
1: Enable the function (Assigned from Fieldbus)

* This function is implemented using the standard ladder program of TCmini (internal PLC).

PLC parameter [P003] settings

Setting for disabling the function (SEQ-FUNCTION-SW1= 0)
The system signal is assigned to [X00W] and [Y00W] of the system IO connector.

```
‘[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 0, 0, 0, 1, 0, 0, 0, 0
```

Setting for enabling the function (SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3 = 0)
The system signal is assigned to [X27W] and [Y37W] of the Fieldbus.

```
‘[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 1, 0, 0, 1, 0, 0, 0, 0
```

Setting for enabling the function (SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3= 1)
The system signal is assigned to [X26W] and [Y36W] of the Fieldbus.

```
‘[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 1, 0, 1, 1, 0, 0, 0, 0
```

Setting for enabling the function (SEQ-FUNCTION-SW2 = 0)
The command mode register of the system signal is assigned to [G17B] of the touch panel communication.

```
‘[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 0, 0, 0, 1, 0, 0, 0, 0
```

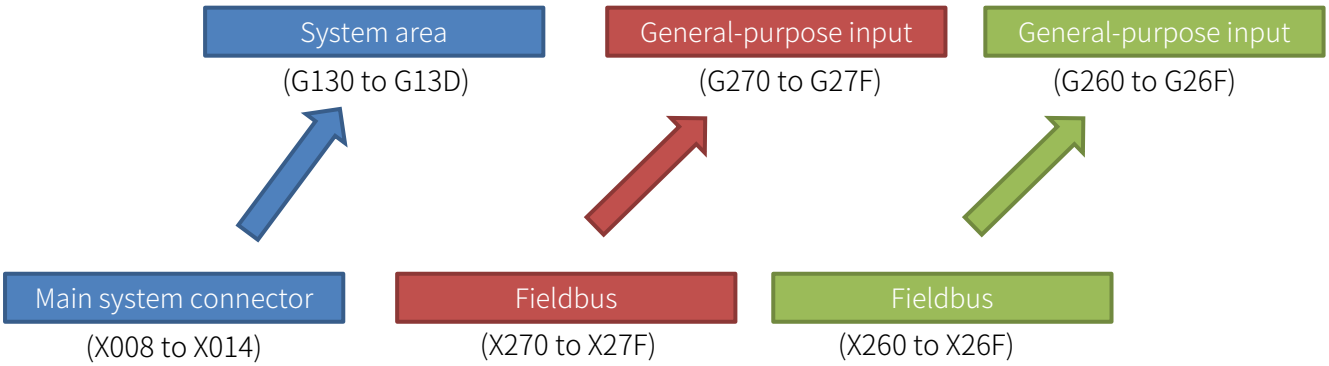
Setting for enabling the function (SEQ-FUNCTION-SW2 = 1)
The command mode register of the system signal is assigned to [X26F] of the Fieldbus.

```
‘[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 0, 1, 0, 1, 0, 0, 0, 0
```

13.1 System Input Signal Assignment Configuration

The input address that will be written changes by function disabled/enabled.

13.1.1 System Configuration at Function Disabled (SEQ-FUNCTION-SW1= 0)



System area	G13F	G13E	G13D	G13C	G13B	G13A	G139	G138	G137	G136	G135	G134	G133	G132	G131	G130
	-	-	BZ_ RST	SV_ OFF	BRE AK	LOW _SPD	CYCLE	STOP	EX_ SVON	RUN	ALM _RST	DO _RST	CYC _RST	STEP _RST	PRG _RST	STR OBE
Main system connector	-	-	-	X010	X011	X012	X013	X014	X00F	X00E	X00D	X00C	X00B	X00A	X009	X008

General-purpose input area	G26F	G26E	G26D	G26C	G26B	G26A	G269	G268	G267	G266	G265	G264	G263	G262	G261	G260
	DIN448	DIN447	DIN446	DIN445	DIN444	DIN443	DIN442	DIN441	DIN440	DIN439	DIN438	DIN437	DIN436	DIN435	DIN434	DIN433
Fieldbus	X26F	X26E	X26D	X26C	X26B	X26A	X269	X268	X267	X266	X265	X264	X263	X262	X261	X260

General-purpose input area	G27F	G27E	G27D	G27C	G27B	G27A	G279	G278	G277	G276	G275	G274	G273	G272	G271	G270
	DIN464	DIN463	DIN462	DIN461	DIN460	DIN459	DIN458	DIN457	DIN456	DIN455	DIN454	DIN453	DIN452	DIN451	DIN450	DIN449
Fieldbus	X27F	X27E	X27D	X27C	X27B	X27A	X279	X278	X277	X276	X275	X274	X273	X272	X271	X270

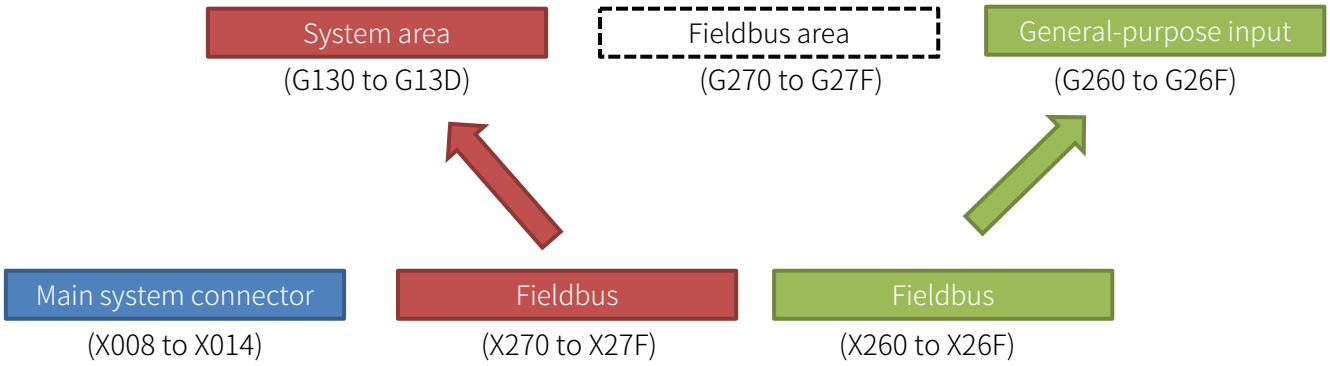
System area	G14F
	DATA_TRIG
Connection signal	* 1 G17B (touch screen input)
	* 2 X26F- (Fieldbus input)

*1 Setting the function to Disable (SEQ-FUNCTION-SW2 = 0)

*2 Setting the function to Enable (SEQ-FUNCTION-SW2 = 1)

13.1.2 System Configuration at Function Enabled

(SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3 = 0)



System area	G13F	G13E	G13D	G13C	G13B	G13A	G139	G138	G137	G136	G135	G134	G133	G132	G131	G130
	-	-	BZ_ RST	SV_ OFF	BRE AK	LOW _SPD	CYCLE	STOP	EX_ SVON	RUN	ALM _RST	DO _RST	CYC _RST	STEP _RST	PRG _RST	STR OBE
Fieldbus	-	-	X27D	X27C	X27B	X27A	X279	X278	X277	X276	X275	X274	X273	X272	X271	X270

General-purpose input area	G26F	G26E	G26D	G26C	G26B	G26A	G269	G268	G267	G266	G265	G264	G263	G262	G261	G260
	DIN448	DIN447	DIN446	DIN445	DIN444	DIN443	DIN442	DIN441	DIN440	DIN439	DIN438	DIN437	DIN436	DIN435	DIN434	DIN433
Fieldbus	X26F	X26E	X26D	X26C	X26B	X26A	X269	X268	X267	X266	X265	X264	X263	X262	X261	X260

General-purpose input area	G27F	G27E	G27D	G27C	G27B	G27A	G279	G278	G277	G276	G275	G274	G273	G272	G271	G270
	DIN464	DIN463	DIN462	DIN461	DIN460	DIN459	DIN458	DIN457	DIN456	DIN455	DIN454	DIN453	DIN452	DIN451	DIN450	DIN449
No assignment	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

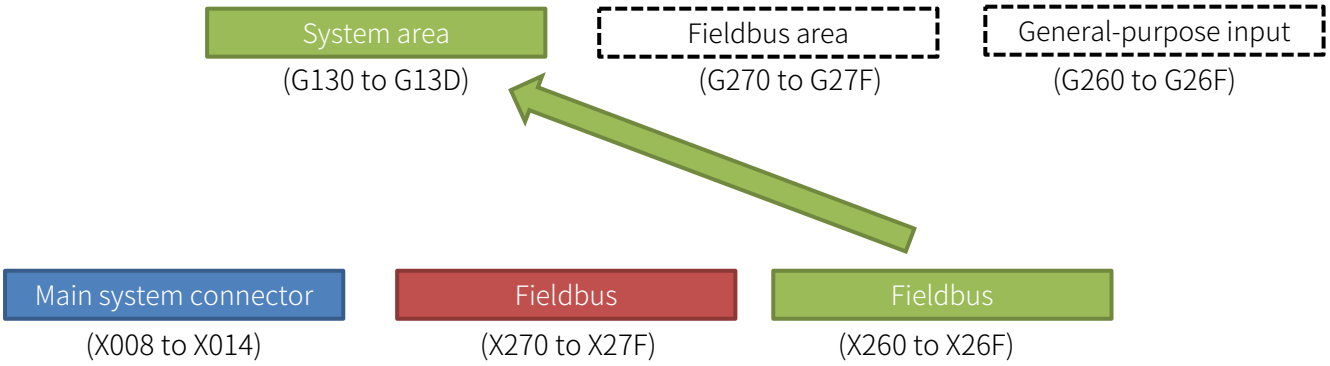
System area	G14F
	DATA_TRIG
Connection signal	* 1 G17B (touch screen input)
	* 2 X26F- (Fieldbus input)

*1 Setting the function to Disable (SEQ-FUNCTION-SW2 = 0)

*2 Setting the function to Enable (SEQ-FUNCTION-SW2 = 1)

13.1.3 System Configuration at Function Enabled

(SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3= 1)



System area	G13F	G13E	G13D	G13C	G13B	G13A	G139	G138	G137	G136	G135	G134	G133	G132	G131	G130
	-	-	BZ_ RST	SV_ OFF	BRE AK	LOW _SPD	CYCLE	STOP	EX_ SVON	RUN	ALM _RST	DO _RST	CYC _RST	STEP _RST	PRG _RST	STR OBE
Fieldbus	-	-	X26D	X26C	X26B	X26A	X269	X268	X267	X266	X265	X264	X263	X262	X261	X260

General-purpose input area	G26F	G26E	G26D	G26C	G26B	G26A	G269	G268	G267	G266	G265	G264	G263	G262	G261	G260
	DIN448	DIN447	DIN446	DIN445	DIN444	DIN443	DIN442	DIN441	DIN440	DIN439	DIN438	DIN437	DIN436	DIN435	DIN434	DIN433
No assignment	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

General-purpose input area	G27F	G27E	G27D	G27C	G27B	G27A	G279	G278	G277	G276	G275	G274	G273	G272	G271	G270
	DIN464	DIN463	DIN462	DIN461	DIN460	DIN459	DIN458	DIN457	DIN456	DIN455	DIN454	DIN453	DIN452	DIN451	DIN450	DIN449
No assignment	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

System area	G14F
	DATA_TRIG
Connection signal	* 1 G17B (touch screen input)
	* 2 X26F- (Fieldbus input)

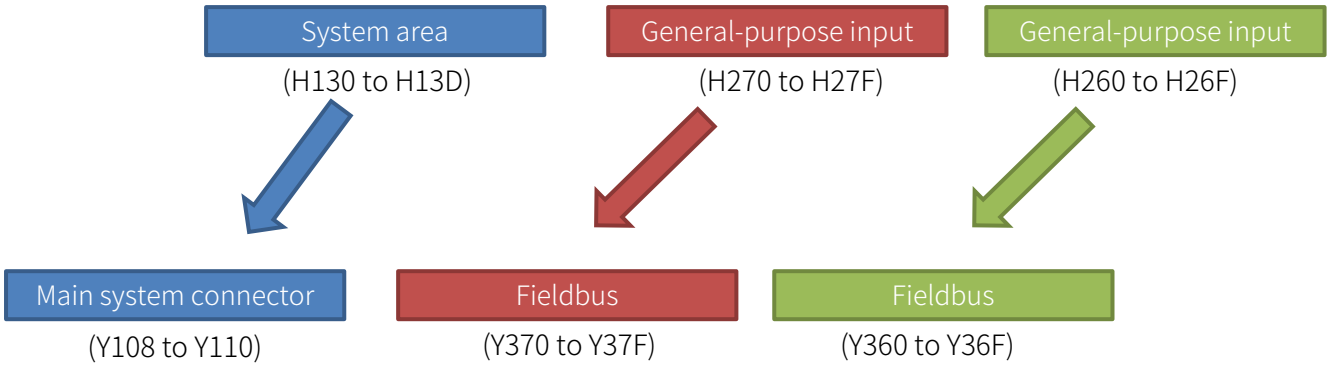
*1 Setting the function to Disable (SEQ-FUNCTION-SW2 = 0)

*2 Setting the function to Enable (SEQ-FUNCTION-SW2 = 1)

13.2 System Output Signal Assignment Configuration

The output address that will be written changes by function disabled/enabled.

13.2.1 System Configuration at Function Disabled (SEQ-FUNCTION-SW1= 0)



System area	H13D	H13C	H13A	H139	H138	H137	H134	H131	H132
	ALARM	BT_ALM	LOW_ST	CYC_END	AUTO_RUN	SYS_RDY	AUTO	SV_RDY	ACK
Main system connector	Y110	Y10F	Y10E	Y10D	Y10C	Y10B	Y10A	Y109	Y108

General-purpose input area	H26F	H26E	H26D	H26C	H26B	H26A	H269	H268	H267	H266	H265	H264	H263	H262	H261	H260
	DOUT448	DOUT447	DOUT446	DOUT445	DOUT444	DOUT443	DOUT442	DOUT441	DOUT440	DOUT439	DOUT438	DOUT437	DOUT436	DOUT435	DOUT434	DOUT433
Fieldbus	Y36F	Y36E	Y36D	Y36C	Y36B	Y36A	Y369	Y368	Y367	Y366	Y365	Y364	Y363	Y362	Y361	Y360

General-purpose input area	H27F	H27E	H27D	H27C	H27B	H27A	H279	H278	H277	H276	H275	H274	H273	H272	H271	H270
	DOUT464	DOUT463	DOUT462	DOUT461	DOUT460	DOUT459	DOUT458	DOUT457	DOUT456	DOUT455	DOUT454	DOUT453	DOUT452	DOUT451	DOUT450	DOUT449
Fieldbus	Y37F	Y37E	Y37D	Y37C	Y37B	Y37A	Y379	Y378	Y377	Y376	Y375	Y374	Y373	Y372	Y371	Y370

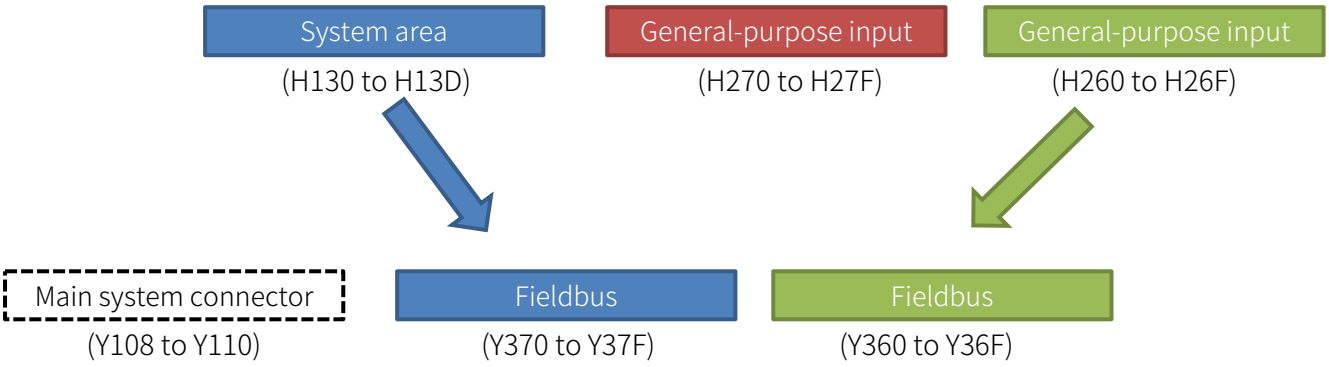
Fieldbus	Y36F	Y36E
Connection signal	* 1 H26F (general-purpose input)	* 1 H26E (general-purpose input)
	* 2 H14F (DATA_ACK)	* 2 H14E (DATA_ERR)

*1 Setting the function to Disable (SEQ-FUNCTION-SW2 = 0)

*2 Setting the function to Enable (SEQ-FUNCTION-SW2 = 1)

13.2.2 System Configuration at Function Enabled

(SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3 = 0)



No assignment	-	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-	-
Main system connector	Y110	Y10F	Y10E	Y10D	Y10C	Y10B	Y10A	Y109	Y108

General-purpose input area	H26F DOUT 448	H26E DOUT 447	H26D DOUT 446	H26C DOUT 445	H26B DOUT 444	H26A DOUT 443	H269 DOUT 442	H268 DOUT 441	H267 DOUT 440	H266 DOUT 439	H265 DOUT 438	H264 DOUT 437	H263 DOUT 436	H262 DOUT 435	H261 DOUT 434	H260 DOUT 433
Fieldbus	Y36F	Y36E	Y36D	Y36C	Y36B	Y36A	Y369	Y368	Y367	Y366	Y365	Y364	Y363	Y362	Y361	Y360

System area	-	-	H13E	H13D	H13C	H13A	H139	H138	H137	H136	H135	H134	H133	H132	H131	H130
	-	-	- *3 ALARM*4	ALARM*3 BT_ALM*4	BT_ALM*3 CYC_ST*4	LOW_ST	CYC_END	AUTO_RUN	SYS_RDY	-	-	AUTO	MANUAL	ACK	SV_RDY	EMG_ST
Fieldbus	Y37F	Y37E	Y37D	Y37C	Y37B	Y37A	Y379	Y378	Y377	Y376	Y375	Y374	Y373	Y372	Y371	Y370

Fieldbus	Y36F	Y36E
Connection signal	* 1 H26F (general-purpose input)	* 1 H26E (general-purpose input)
	* 2 H14F (DATA_ACK)	* 2 H14E (DATA_ERR)

*1 Setting the function to Disable (SEQ-FUNCTION-SW2 = 0)

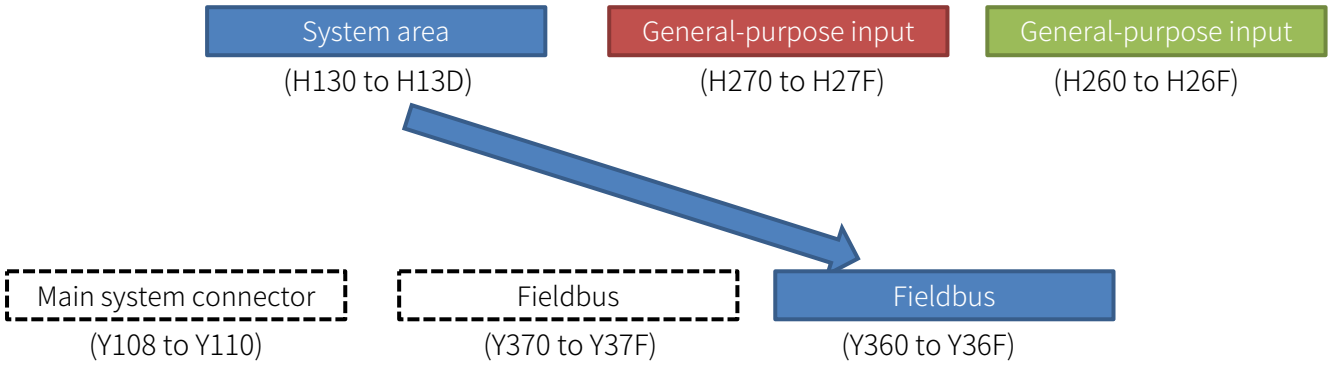
*2 Setting the function to Enable (SEQ-FUNCTION-SW2 = 1)

*3 Setting the function to Disable (SEQ-FUNCTION-SW4 = 0)

*4 Setting the function to Enable (SEQ-FUNCTION-SW4 = 1)

13.2.3 System Configuration at Function Enabled

(SEQ-FUNCTION-SW1 = 1, SEQ-FUNCTION-SW3= 1)



No assignment	-	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-	-
Main system connector	Y110	Y10F	Y10E	Y10D	Y10C	Y10B	Y10A	Y109	Y108

System area	-	-	H13E	H13D	H13C	H13A	H139	H138	H137	H136	H135	H134	H133	H132	H131	H130
	-	-	- *3 ALA RM*4	ALA RM*3 BT_ ALM*4	BT_ ALM*3 CYC_ST *4	LOW_ ST	CYC_ END	AUTO_ RUN	SYS_ RDY	-	-	AUTO	MANUAL	ACK	SV_ RDY	EMG _ST
Fieldbus	Y36F	Y36E	Y36D	Y36C	Y36B	Y36A	Y369	Y368	Y367	Y366	Y365	Y364	Y363	Y362	Y361	Y360

No assignment	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fieldbus	Y37F	Y37E	Y37D	Y37C	Y37B	Y37A	Y379	Y378	Y377	Y376	Y375	Y374	Y373	Y372	Y371	Y370

Fieldbus	Y36F	Y36E
Connection signal	* 1 H26F (general-purpose input)	* 1 H26E (general-purpose input)
	* 2 H14F (DATA_ACK)	* 2 H14E (DATA_ERR)

*1 Setting the function to Disable (SEQ-FUNCTION-SW2 = 0)

*2 Setting the function to Enable (SEQ-FUNCTION-SW2 = 1)

*3 Setting the function to Disable (SEQ-FUNCTION-SW4 = 0)

*4 Setting the function to Enable (SEQ-FUNCTION-SW4 = 1)

14. TCmini (Internal PLC) Function (Sample Connection Example)

The setting method when using an upper-level PLC (such as the KV-7500 by Keyence) to control the controller by Ethernet/IP specifications is shown below. These settings assume that the following settings have already been made.

[***] Address of TCmini (internal PLC) in controller
[X20W to X27W]: Fieldbus input address
[X40W to X47W]
[X60W to X67W]
[X80W to X87W]
[Y30W to Y37W]: Fieldbus output address
[Y50W to Y57W]
[Y70W to Y77W]
[Y90W to Y97W]
[D700 to D70F]: Data input address
[D740 to D74F]: Data output address
[D460 to D46F]: Data command address
[D660 to D66F]: Data response address
[W000 to W05F]: General-purpose input/output address in KV-7500 by Keyence

■ Upper-level PLC assignment setting example
Fieldbus General-purpose I/O Section
The controller Fieldbus output is assigned to the input of the KY-7500 by Keyence.
[Y30W to Y37W] → [W000 to W007] 128 each
[Y50W to Y57W] → [W008 to W00F] 128 each
[Y70W to Y77W] → [W010 to W017] 128 each
[Y90W to Y97W] → [W018 to W01F] 128 each

Output of the KV-7500 by Keyence is assigned to the input of the controller Fieldbus input.
[X20W to X27W] → [W030 to W037] 128 each
[X40W to X47W] → [W038 to W03F] 128 each
[X60W to X67W] → [W040 to W047] 128 each
[X80W to X87W] → [W048 to W04F] 128 each

Fieldbus PLC Data Section
The controller Fieldbus data output is assigned to the input of the KY-7500 by Keyence.
[D740 to D74F] → [W020 to W02F]

Output of the KV-7500 by Keyence is assigned to the input of the controller Fieldbus input.
[D700 to D70F] → [W050 to W05F]

The addresses [W000 to W05F] to be assigned vary depending on the manufacturer of the connected upper-level PLC.

14.1 Operating the Robot from an Upper-level PLC

(1) Set the value in row 1 column 1 (FUNCTION 1) for [P003] SEQUENCE FUNCTION SELECT SWITCH to 1, and assign the system signal input/output to the Fieldbus. (Parameter enabled status)

(2) Check whether the controller information is recognized in the upper-level PLC. Conversely, write the value from the master device, and check that the signal is activated.

Example: Check whether the stop signal of the controller can be controlled from the master device.

- Set the 8th bit of the master device address [W01F] to 1, press the buttons UTILITY - [I/O] - [SYS] - NEXT on the teach pendant so that the following screen appears.
- Check whether 1 is set to the STOP item on the teach pendant screen.

Upper-level PLC output address

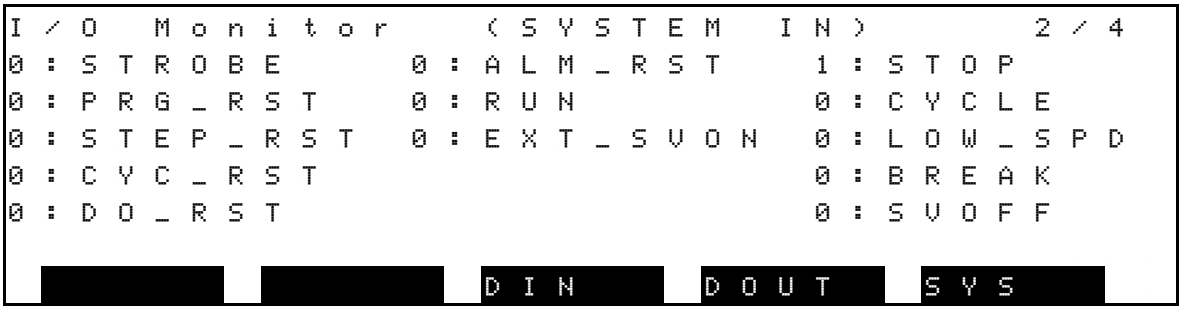
Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[W037]	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

↓

Input address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[X27W]				SV OFF	BREAK	LOW_ SPD	CYC LE	STOP	EX_ SVON	RUN	ALM _RST	DO _RST	CYC _RST	STEP _RST	PRG_ RST	STRO BE

Teach pendant screen (TP1000)

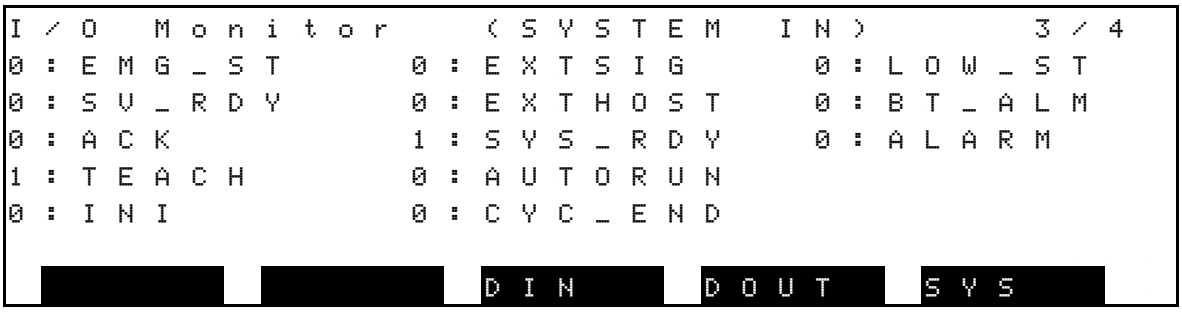


- Next, check that the value in the controller status changes by the master device.
  When checking, the controller operation mode must be AUTO mode.

- Press the buttons UTILITY - [I/O] - [SYS] - NEXT twice on the teach pendant so that the following screen appears.

- Check that the value of the master device input address changes when the teach pendant is used to change the mode from MANUAL to AUTO.

Teach pendant screen (TP1000)



Output address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[Y37W]			FO126	ALARM	BT_ ALM	LOW _ST	CYC_ END	AUTO RUN	SYS_ RDY	FO119	FO118	AUTO	MANU AL	ACK	SV_ RDY	EMG_ ST

↓

Master device input address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[W007]	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1

- Check that 1 is set to the 0th, 4th, and 7th bit of the upper-level PLC [W007] address.

Input the EX_SVON signal from the master device.

- Press the ESC button until the default screen [EDIT] appears on the teach pendant screen.
- Set the controller operation mode key to AUTO mode.
- Press the ERR button on the teach pendant to check that there are no alarms. (The emergency stop switch is cleared.)
- Ensure that the area around the robot is safe for operation.
- First, set 1 to the SVOFF signal, and then input the command from the upper-level PLC so that the EX_SVON signal becomes a pulse signal (100-msec width).

Upper-level PLC output address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[W037]	0	0	0	1	0	0	0	0	0→1 →0	0	0	0	0	0	0	0

↓

Input address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[X27W]				SV OFF	BREAK	LOW_ SPD	CYC LE	STOP	EX_ SVON	RUN	ALM_ RST	DO_ RST	CYC_ RST	STEP _RST	PRG_ RST	STRO BE

- Check that the Servo ON switch on the teach pendant is lit green. Also, check that 1 is set to the 2nd bit (SV_RDY) of the master device address [W07].

Output address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[Y37W]			FO126	ALARM	BT_ ALM	LOW_ ST	CYC_ END	AUTO RUN	SYS_ RDY	FO119	FO118	AUTO	MANU AL	ACK	SV_ RDY	EMG_ ST

↓

Upper-level PLC input address

Bit	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
[W007]	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1

- Set the SVOFF signal to 0, and check that the Servo ON switch on the teach pendant turns off. Also, check that 0 is set to the 1st bit (SV_RDY) of the master device address [W027].

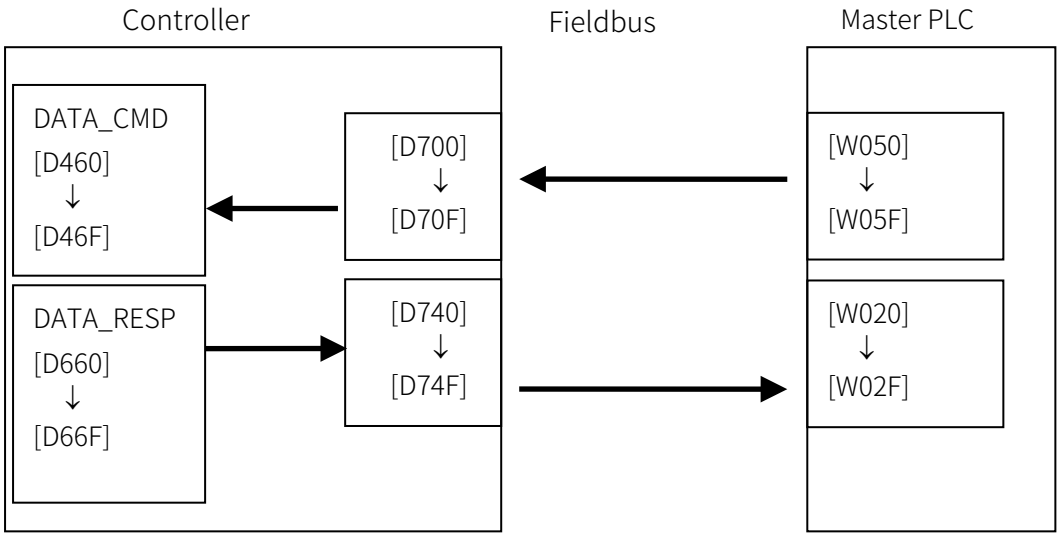
MEMO

- The configuration does not allow Servo On for about 4.5 seconds after Servo Off to prevent internal damage. - Execute the EX_SVON signal again after at least 5 seconds have elapsed after Servo Off. Also, when Servo On is performed for the 3rd time, the interval time becomes 7 seconds.

14.2 Sending Robot Position Information to Master Device

Commands can be sent from the master device to obtain the robot joint coordinate positions, world coordinate current positions, and work coordinate current positions. This data can be sent to the master device.

Schematic Diagram



Fieldbus Data Communication Function

Set the value in row 1 column 2 (FUNCTION 2) for the PLC parameter [P003] SEQUENCE FUNCTION SELECT SWITCH to 1, and restart the controller to enable the function.

'[P003] SEQUENCE FUNCTION SELECT SWITCH SEQ-FUNCTION-SW = 0, 1, 0, 0, 0, 0, 0, 0

When this function is enabled, the settings are as follows.

H14E: Command error signal → Assignment to Fieldbus output (FO111)

H14F: Response complete signal → Assignment to Fieldbus output (FO112)

G14F: Request trigger signal → Assignment to robot program "DIN(448)"

D70*: Fieldbus data input register → Data is transferred to D46* (Command set register)

D74*: Fieldbus data output register → Data is received from D66* (Command response register)

Command is sent from the master device.

Example: The world coordinate current position commands of monitor mode are sent.

The master device address is set to [W050]: 0xE311, [W051]: 0x0000.

Upper-level PLC output address

[W05*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E311

↓

Data input address

[D70*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E311

↓

DATA_CMD input address

[D46*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value															0	0x E311

The world coordinate current position is obtained from [W020] to [W02F].

[D66*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			World T coordinate current position		World C coordinate current position		World Z coordinate current position		World Y coordinate current position		World X coordinate current position		Ori-enta-tion		0	0x E311
6-axis	World C coordinate current position		World B coordinate current position		World A coordinate current position											

↓

Data output address

[D74*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value	**	**	**	**	**	**	**	**	**	**	**	**	**	**	0	0x E311

↓

Upper-level PLC input address

[W02*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value	**	**	**	**	**	**	**	**	**	**	**	**	**	**	0	0x E311

14.3 Sending Teaching Points from an Upper-level PLC to Robot

The PLC data communication function is used.

The teaching point data writing command is sent.

> *MEMO*
>
> *- The operation sequence of the command modes are described in chapter 8 of the STA1168_TCmini (Internal PLC) Manual.*

Example: The teaching point data writing command of the command mode is sent.
Any selected teaching point data is set to the PLC addresses [W050]: 0xC2C1, [W051]: 0x0000, and [W052 to W05F].

Upper-level PLC output address

[W05*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value	***	***	***	***	***	***	***	***	***	***	***	***	***	0	Table number	0x C2C1

↓

Data input address

[D70*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value	***	***	***	***	***	***	***	***	***	***	***	***	***	0	Table number	0x C2C1

↓

DATA_CMD input address

[D46*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
SCARA			Teaching point data T-coordinate value		Teaching point data C-coordinate value		Teaching point data Z-coordinate value		Teaching point data Y-coordinate value		Teaching point data X-coordinate value		Orien-tation	0	Table number	0x C2C1
6-axis	Teaching point data C-coordinate value		Teaching point data B-coordinate value		Teaching point data A-coordinate value											

The status is input in [W020 to W02F] for checking that the values were set correctly.

DATA_RESP output address

[D66*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value													0	Error code	Table number	0x C2C1

↓

Data output address

[D74*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value													0	Error code	Table number	0x C2C1

↓

Upper-level PLC input address 20

[W02*]	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
Value													0	Error code	Table number	0x C2C1

MEMO

- The table number indicates the number of the memory variable (POINT type). The value saved to the memory variable (POINT type) is retained even after the power is turned off. For details, refer to the STA1193_Robot Language Manual (SCOL2).

- If no value is saved to the memory variable, an indefinite value is stored to the memory variable.

15. Operations Using SCOL2 Language

The Fieldbus signals can be controlled using commands in SCOL2 language.
For detailed instructions on how to use SCOL2 language commands, refer to the *STA1193 Robot Language Manual (SCOL2)*.

DIN	:	Loads the status of the specified input signal.
DOUT	:	Outputs the specified output signal.
RESET DOUT	:	Turns off the user output signal.
PULOUT	:	Outputs the specified output signal as a 0.2-second width pulse. The output time by the PULOUT command to the above output area may be extended from 0.2 seconds, which is the normal DOUT output time.
BCDIN	:	Loads an input signal with 4x signal length from the signal name as a BCD code.
BCDOUT	:	Converts the value in the formula to a BCD code and outputs the number of digits specified in the signal length from the signal name to an output signal with 4x signal length.
HEXIN	:	Loads an input signal with signal length from the signal name as a HEX code.
HEXOUT	:	Converts the value in the formula to a HEX code and outputs the number of digits specified in the signal length to an output signal with the signal length from the signal name.

15.1 Sample Program

In this program, operation waits until the input signal 301 is set to ON, and then movement is performed to the points A1, A2, and A3 that were taught.

```
PROGRAM    DINSAMPLE
WAIT DIN(301)
MOVE A1
MOVE A2
MOVE A3
END
DATA
POINT A3      =   1500.000,      0.000,      0.000,      0.000,      0.000 / RIGHTY
POINT A2      =   1500.000,      0.000,      0.000,      0.000,      0.000 / RIGHTY
POINT A1      =   1500.000,      0.000,      0.000,      0.000,      0.000 / RIGHTY
END
```

This program sets the output signals 301 to 316 to ON and OFF in sequential order.

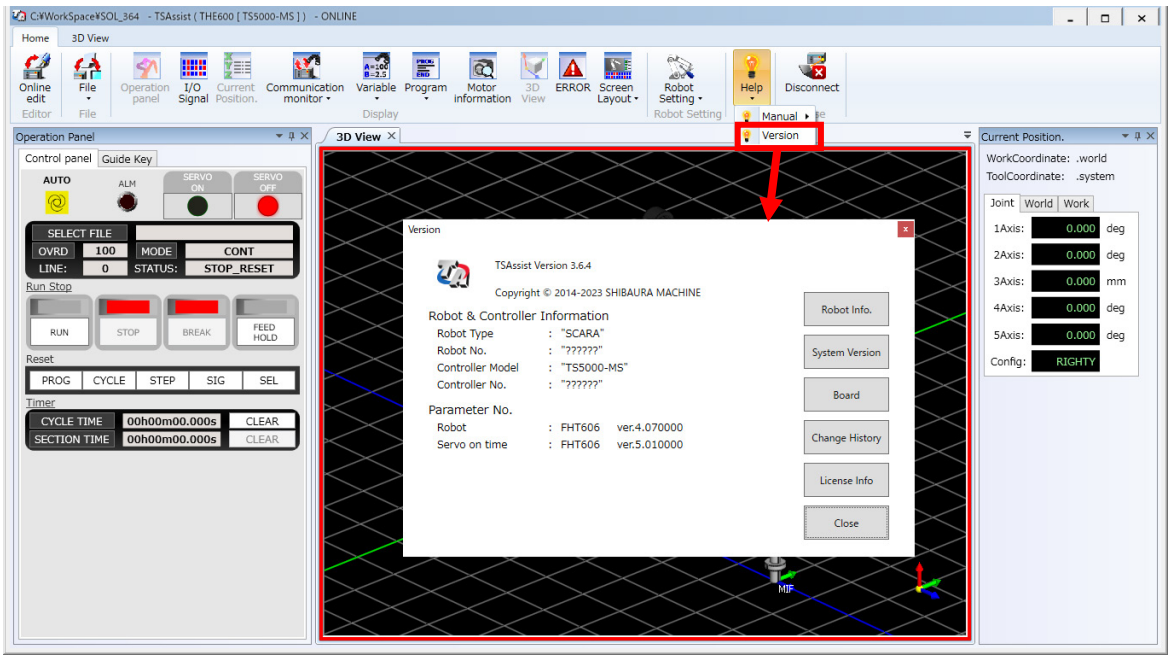
```
PROGRAM    DOUTSAMPLE
FOR K=301 TO 316
DOUT(K)
TIMER=0.5
WAIT TIMER==0
DOUT(-K)
NEXT K
END
```

16. Verifying the Module Version

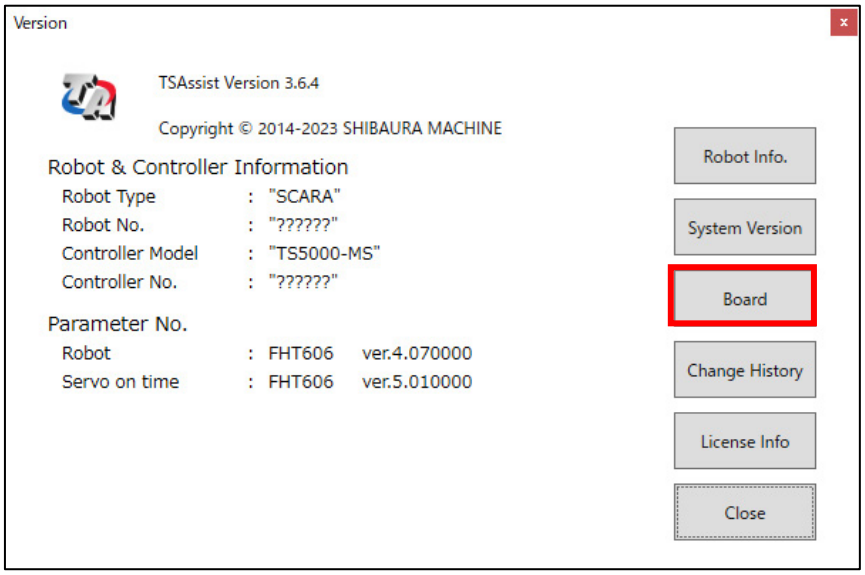
The version of the Fieldbus module can be verified by using TSAssist or TP5000.

16.1 Verifying the Version of the Fieldbus by Using TSAssist

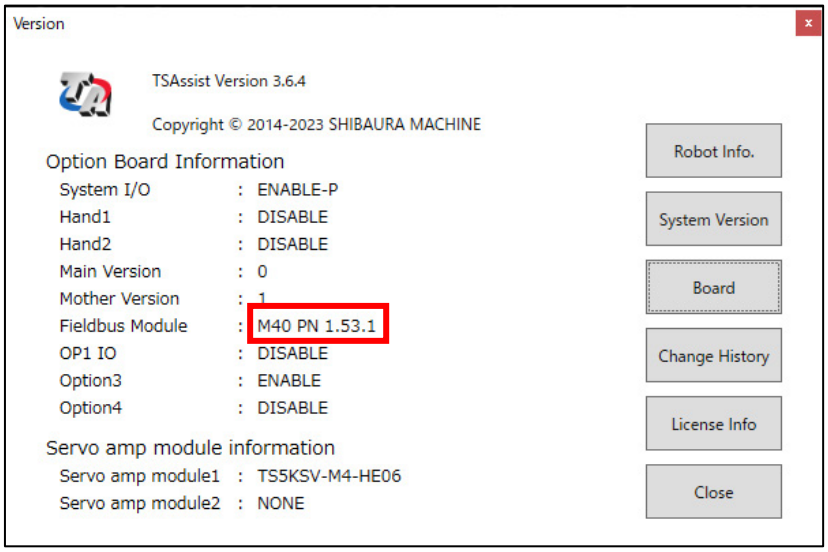
① Click on [Help]-[Version] on the ribbon menu. A version screen is displayed.



② Click on the [Board] button on the Version screen. Board information is displayed.



③ Verify the version that is displayed in [Fieldbus Module].

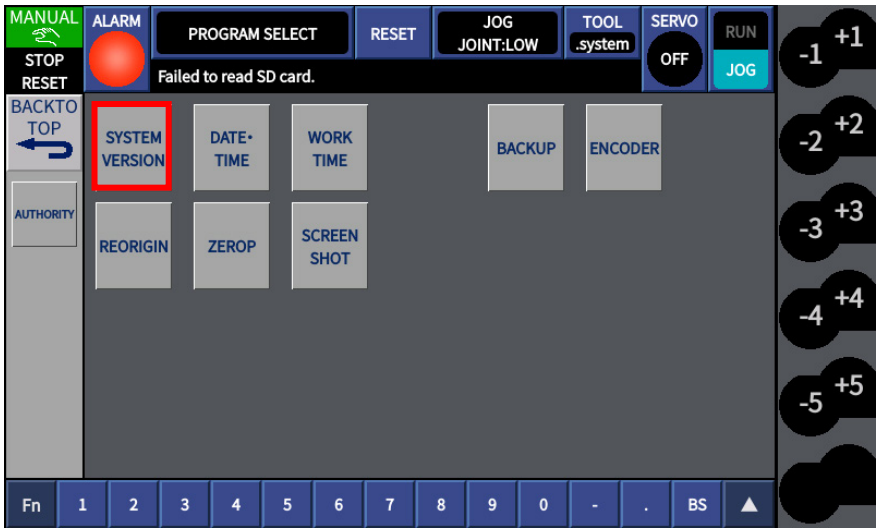


16.2 Verifying the Version of the Fieldbus by using TP5000

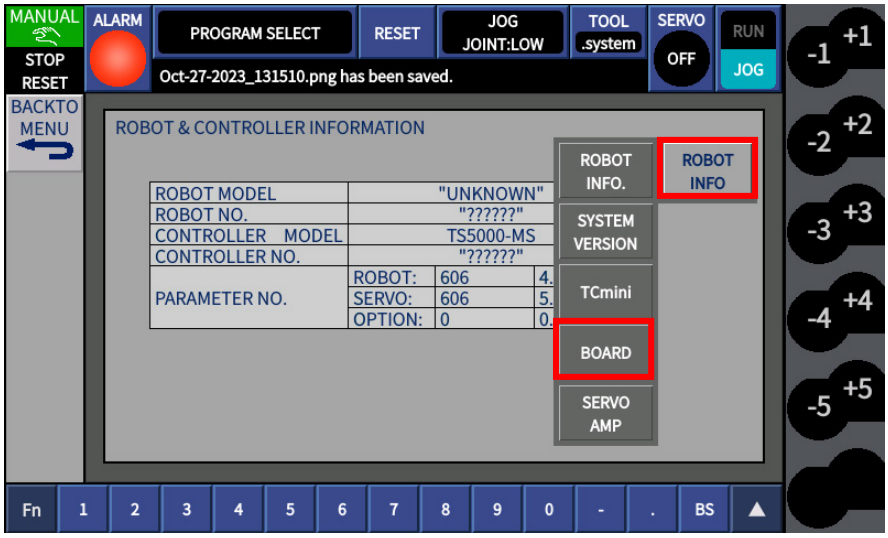
① Tap on the [MAINTE] button. A maintenance menu screen displayed.



② Tap on the [SYSTEM VERSION] button. A version information screen is displayed.



③ Tap on the [ROBOT INFO] button. A [Robot information switching] menu is displayed. Tap on the [BOARD] button on the [Robot information switching] menu.



④ Verify the version that is displayed in [Fieldbus Module].

